# Doubly Robust Nonparametric Local Projections

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**Abstract**

Nonparametric local projections estimate average impulse responses to structural shocks without imposing parametric assumptions on the response function. This paper shows that the same average response can alternatively be identified by reweighting outcomes according to how a structural shock shifts the shock density. Combining both representations yields a doubly robust estimator: a nonparametric regression estimate augmented with a residual correction based on shock-density reweighting. Consistency requires only that either the outcome regression or the density ratio be consistently estimated, making the method less vulnerable to smoothing and specification errors. To accommodate serial dependence, I use buffer-separated block cross-fitting, product-rate nuisance conditions, and HAC inference based on the long-run variance of the orthogonal score. Monte Carlo evidence highlights a practical inferential tradeoff: the residual correction removes persistent regression bias and substantially improves empirical coverage of HAC confidence intervals relative to regression-only estimation. When the regression arm is already accurate, the correction can add residual-weight variation and increase RMSE, especially for large shock shifts, but the resulting intervals are generally better calibrated at the same smoothing choices.

**JEL Codes:** C14, C22, C32
**Keywords:** local projections; double robustness; orthogonal estimation; impulse responses

*Computations were performed using the Wake Forest University High Performance Computing Facility. All errors are my own.

# 1 Introduction

Impulse responses can be highly nonlinear: structural shocks may generate asymmetric, state-dependent, or localized propagation patterns that linear local projections (Jordà, 2005) and vector autoregressions obscure. Nonparametric local projections recover such responses without imposing functional-form restrictions on how shocks propagate (Gonçalves et al., 2024a). But their flexibility introduces a vulnerability — each response estimate rests on a nonparametric regression of future outcomes on the structural shock. When that regression smooths over sharp features of the conditional mean, or when the chosen nuisance function class fails to approximate them, the estimated impulse response inherits the resulting bias. This paper develops a doubly robust estimator for nonparametric local projections that offers insurance against this regression-induced bias.

I show that the impulse response function of Gonçalves et al. (2024a) admits not only a conditional-regression representation but also a reweighting representation. Shifting the shock distribution by a fixed amount is equivalent to reweighting observations by the ratio of the shifted to the original marginal shock density. Combining the two representations yields an orthogonal, doubly robust estimating moment: a regression estimate augmented with a density-ratio-weighted residual correction. The moment is locally insensitive to first-stage errors in the conditional mean and density ratio, so small errors in either component affect the estimator only through their product. Its expectation is zero if either the conditional mean or the density ratio is correctly specified, given mild envelope conditions on the other.

The combined representation leads directly to a feasible estimator. At each horizon and shock displacement, the estimator averages over time a regression component and a density-ratio-weighted residual correction. The only unknown ingredients are the conditional mean and the density ratio, both of which may be estimated flexibly. Orthogonality of the moment ensures that first-stage errors enter only through a product term, making them asymptotically negligible under a product-rate condition. I establish consistency and asymptotic normality, with inference based on a HAC estimate of the long-run variance. Monte Carlo simulations show that the estimator reduces the persistent smoothing and approximation bias of regression-only nonparametric local projections and improves the empirical coverage of nominal HAC intervals. The coverage evidence is important because HAC standard errors measure stochastic score variation but do not remove deterministic smoothing or specification bias; a regression-only interval can therefore undercover even when its point bias appears small on the scale of the response. The DR score attenuates this centering error through the product-of-errors

remainder, at the cost of additional residual-weight variation that can raise RMSE when the regression arm is already accurate.

The paper connects three literatures: nonlinear impulse response analysis, counterfactual density-ratio representations, and semiparametric orthogonal estimation. Gonçalves et al. (2024a) formulate nonlinear impulse responses as potential-outcome objects and estimate them by nonparametric local projections; this paper takes their framework as its starting point. The reweighting representation links the paper to work on stochastic interventions and modified treatment policies, where counterfactual distributions are generated by shifting a continuous exposure rather than by setting a binary treatment (Díaz Muñoz and van der Laan, 2012; Haneuse and Rotnitzky, 2013; Díaz et al., 2023). It also connects to density-ratio and importance-weighting methods, which estimate expectations under one distribution by reweighting observations drawn from another (Sugiyama et al., 2012). The treatment-effects literature developed doubly robust estimation for both binary and continuous treatments (Robins et al., 1994; Hirano and Imbens, 2004; Kennedy et al., 2017), along with sample-splitting methods for flexible nuisance estimation (Chernozhukov et al., 2018).

This paper makes three contributions. First, I show that the impulse response function admits two distinct observable representations rather than only one. Existing nonparametric local projections recover the response entirely through a regression of future outcomes on the structural shock. That regression must be re-estimated at each forecast horizon, and the response inherits whatever smoothing or approximation bias it carries. I show that the same response can also be recovered by reweighting observations according to how the shock distribution shifts. The reweighting component depends only on the marginal density of the shock, so it does not change with the forecast horizon and is lower-dimensional than the outcome regression. These properties make it a natural complement: the horizon-specific part of the problem is isolated in the regression, and the stable, low-dimensional part is isolated in the density ratio. This decomposition provides the second arm needed for doubly robust estimation.

Second, I combine these two representations into a doubly robust estimator of the impulse response function. The estimator augments a regression estimate with a residual correction based on shock-density reweighting. Its key property is that the impulse response is consistently estimated whenever either the outcome regression or the density ratio is consistently estimated. If the regression is accurate, the augmentation is innocuous. If the regression is biased, the density-ratio correction removes that bias. This gives the researcher insurance against the bandwidth choices, series orders, and function-class decisions that govern the regression step. I formalize this protection

through Neyman orthogonality: first-stage estimation errors affect the estimator only through their product rather than individually.

Third, I develop the inference theory needed to deploy this estimator in time-series settings. Standard cross-fitting procedures for orthogonal estimators assume independent observations. Impulse response data are serially dependent, so those procedures do not directly apply. I introduce a block cross-fitting scheme that trains each nuisance function on one temporal block and evaluates it on a separate block, with a buffer between them to attenuate dependence across the training and evaluation sets. Under a product-rate condition on the two nuisance estimation errors, the estimator is asymptotically normal. I estimate its long-run variance with a HAC estimator that accounts for residual dependence in the orthogonal score sequence.

**Related Literature.** This paper builds on the local projection framework introduced by Jordà (2005), which estimates impulse responses through horizon-specific regressions rather than VAR cross-equation restrictions. Linear local projections impose proportional and sign-symmetric responses. A large subsequent literature relaxes these restrictions using state interactions, smooth-transition specifications, nonlinear transformations, sign- and size-dependent responses, and threshold regimes (Auerbach and Gorodnichenko, 2012, 2013; Gonçalves et al., 2024b, 2021; Barnichon and Matthes, 2018; Ramey and Zubairy, 2018). These extensions enrich the projection while remaining finite-dimensional: the relevant nonlinearities must be specified by the researcher. Gonçalves et al. (2024a) depart from this paradigm by identifying average impulse responses from the unrestricted conditional mean of future outcomes given the structural shock, which they estimate nonparametrically. My paper takes their estimand and identification result as its starting point but addresses a vulnerability of purely regression-based implementation: any smoothing, bandwidth, sieve, or function-class error in the estimated conditional mean passes directly through to the impulse response. The response therefore inherits the bias–variance properties of a single nonparametric regression, including slower-than-parametric convergence and potentially persistent approximation bias.

The estimator is also related to doubly robust and orthogonal methods in semiparametric causal inference. Augmented inverse propensity weighted estimators combine an outcome regression with an assignment or weighting model and remain consistent when either component is correctly specified (Robins et al., 1994; Bang and Robins, 2005); targeted learning and double/debiased machine learning extend this logic to data-adaptive first stages through orthogonalized scores and sample splitting (van der

Laan and Rose, 2011; Chernozhukov et al., 2018). The closest formal connection is to the Riesz representer approach to debiasing regression functionals (Chernozhukov et al., 2022a,b). In the present setting, the average impulse response is a functional of the conditional mean, and shifting the structural shock supplies the corresponding density ratio adjustment. Related continuous treatment estimators use outcome regressions together with conditional densities or generalized propensity scores to estimate dose-response functions (Kennedy et al., 2017; Colangelo and Lee, 2026). Here the structural shock design changes the nuisance object: because the shock is independent of the other innovations, the relevant weight is the marginal shock density ratio rather than a conditional treatment density.

Finally, the paper connects to orthogonal learners for heterogeneous causal effects (Wager and Athey, 2018; Athey et al., 2019; Nie and Wager, 2021; Kennedy, 2023; Oprescu et al., 2019). These methods use orthogonalization to estimate treatment effect objects under independent sampling with outcome and propensity nuisances. The impulse response setting of this paper differs in three respects: the target is a nonparametric impulse response to a shifted structural shock, the weighting nuisance is a change of measure density ratio rather than a propensity score, and the orthogonal score is serially dependent. These distinctions motivate buffer separated block cross-fitting and HAC inference based on the long run variance of the score.

**Outline.** Section 2 presents the structural model and the conditional-mean/density-ratio identification of the average response function. Section 3 develops the block cross-fitted doubly robust estimator, states its double-robustness and asymptotic properties, and records the influence score and the long-run variance target for HAC inference. Section 4 reports Monte Carlo simulations, with figures collected in Appendix D. Formal assumptions, the influence-function derivation, a sufficient primitive rate verification, and implementation details are collected in the appendices.

## 2 Setup and Identification

An impulse response function characterizes the average effect on outcomes at horizon $h$ of a counterfactual shift of size $\delta$ in the structural shock at time $t$. Defining this object requires three elements: a structural model that isolates the shock of interest from the remaining sources of variation, a potential-outcomes framework that gives the counterfactual precise meaning, and an identification argument that expresses the counterfactual in terms of observable quantities. This section introduces each element in

turn and establishes that identification admits two distinct observable representations — one through the conditional mean of the outcome given the shock, the other through the marginal density of the shock. The interplay between these two representations underlies the doubly robust estimator developed in Section 3.

## 2.1 Structural Model

Consider a vector of observables $z_t = (x_t, Y_t')'$ generated by the structural nonlinear dynamic system

$$x_t = \varphi(\mathbf{z}_{t-1}) + \varepsilon_{1t} \tag{1}$$

$$y_{it} = \psi_i(x_t, Y_{-i,t}, \mathbf{z}_{t-1}, \varepsilon_{it}), \quad i = 2, \ldots, n \tag{2}$$

where $\varepsilon_t = (\varepsilon_{1t}, \ldots, \varepsilon_{nt})'$ is i.i.d. with mean zero and mutually independent components, and boldface denotes the full lag history of a variable. The first equation isolates a single structural shock $\varepsilon_{1t}$, which enters additively; the remaining equations permit the outcomes to depend on the shock, on one another, and on the full history through unrestricted nonlinear functions $\psi_i$. This is the nonlinear structural local projection environment of Gonçalves et al. (2024a), which nests models with nonlinearly transformed regressors (Gonçalves et al., 2021), state-dependent coefficients (Gonçalves et al., 2024b), and nonlinear interactions between shocks and state variables.

Throughout, I treat the structural shock series $\varepsilon_{1t}$ as observed, either directly or as the output of a first-stage identification procedure. All estimators below are written in terms of this series. This is the natural starting point when an external instrument or narrative identification strategy delivers a shock series. Identification rests on a single feature of the specification: the mutual independence of the components of $\varepsilon_t$, which ensures that $\varepsilon_{1t}$ is independent of the remaining innovations.

## 2.2 Potential Outcomes and Estimands

The structural model induces a mapping from the shock $\varepsilon_{1t}$ to future outcomes. To formalize the counterfactual of setting $\varepsilon_{1t}$ to a value $e$ other than its realization, define the potential outcome

$$y_{t+h}(e) \;=\; m_h(e,\, U_{t+h}), \tag{3}$$

where $U_{t+h}$ collects all remaining sources of variation in $y_{t+h}$: lagged states, future shocks $\varepsilon_{1,t+1}, \ldots, \varepsilon_{1,t+h}$, and non-shock innovations $\varepsilon_{2,t}, \ldots, \varepsilon_{n,t+h}$. The function $m_h$ is

determined by the structural equations (1)–(2) but is left unrestricted — no functional form is imposed on how the shock propagates. This is the potential-outcomes formulation employed in nonlinear local projection analysis by Gonçalves et al. (2021, 2024b,a).

The target estimand is the unconditional average response function:

$$\mathrm{ARF}_h(\delta) \;\equiv\; E\big[y_{t+h}(\varepsilon_{1t}+\delta) - y_{t+h}(\varepsilon_{1t})\big]. \tag{4}$$

Following Gonçalves et al. (2021, 2024a), this quantity is a nonparametric generalization of the linear impulse response: it measures the average change in the outcome when every shock realization is shifted by $\delta$, without restricting the response to be linear or symmetric in $\delta$. The fixed-shock response of Koop et al. (1996), which compares $y_{t+h}(e+\delta)$ and $y_{t+h}(e)$ at a chosen baseline $e$, is a distinct object in nonlinear models.

## 2.3 Identification

The ARF involves the potential outcome $y_{t+h}(e)$, which is observed only at the realized shock value $e = \varepsilon_{1t}$. Identification therefore requires a link between the counterfactual and observable quantities. The structural model supplies this link through a single condition: the shock $\varepsilon_{1t}$ is independent of all other determinants of $y_{t+h}$. Formally, $\varepsilon_{1t} \perp\!\!\!\perp U_{t+h}$, which follows from the i.i.d. assumption on $\varepsilon_t$ and the triangular timing in equations (1)–(2): because $\varepsilon_{1t}$ is serially independent and contemporaneously independent of $\varepsilon_{2t}, \ldots, \varepsilon_{nt}$, it is independent of the collection $U_{t+h}$. This in turn implies that the potential outcomes $\{y_{t+h}(e) : e \in \mathcal{E}\}$ are independent of $\varepsilon_{1t}$ — the analog of unconfoundedness in the treatment effects literature; see Lemma A.1 of Gonçalves et al. (2024b) and Gonçalves et al. (2024a). This independence condition opens two representations of the same estimand.

**Representation I: Conditional means.** Define the conditional mean function

$$g_h(e) \;\equiv\; E[y_{t+h} \mid \varepsilon_{1t} = e]. \tag{5}$$

Because $\varepsilon_{1t} \perp\!\!\!\perp U_{t+h}$, the conditional expectation of the potential outcome at any shock value $e$ equals $g_h(e)$:

$$E[y_{t+h}(e)] \;=\; E[m_h(e, U_{t+h})] \;=\; E[y_{t+h} \mid \varepsilon_{1t} = e] \;=\; g_h(e),$$

where the first equality integrates out $U_{t+h}$ using independence, and the second conditions on $\varepsilon_{1t} = e$ in the observed outcome $y_{t+h} = m_h(\varepsilon_{1t}, U_{t+h})$. Substituting into the ARF definition yields the conditional-mean representation:

$$\mathrm{ARF}_h(\delta) \;=\; E\big[g_h(\varepsilon_{1t}+\delta) - g_h(\varepsilon_{1t})\big]. \tag{CM}$$

The impulse response is thus identified by the conditional mean of $y_{t+h}$ given $\varepsilon_{1t}$: estimate $g_h$, evaluate it at the observed and shifted shock values, and average the difference. This identification result and the estimator it motivates are due to Gonçalves et al. (2024a); see their Proposition 4.1 and Algorithm 5.1.

**Representation II: Density ratios.** Rather than expressing the ARF through the conditional mean, one can express it through the marginal density of the shock. The key observation is that shifting every shock realization by $\delta$ is equivalent to reweighting observations under the original shock distribution. Realizations that become more likely under the shifted distribution receive greater weight, and those that become less likely receive less. Let $f$ denote the marginal density of $\varepsilon_{1t}$. A change of variables gives

$$\begin{aligned} E[g_h(\varepsilon_{1t}+\delta)] &= \int g_h(e+\delta)\, f(e)\, de \\ &= \int g_h(u)\, f(u-\delta)\, du \\ &= E\left[g_h(\varepsilon_{1t}) \cdot \frac{f(\varepsilon_{1t}-\delta)}{f(\varepsilon_{1t})}\right]. \end{aligned} \tag{6}$$

Define the density ratio

$$r_\delta(e) \;\equiv\; \frac{f(e-\delta)}{f(e)}. \tag{7}$$

This ratio is well-defined provided the support of $f(\cdot - \delta)$ is contained in that of $f$, an overlap condition that holds whenever $\varepsilon_{1t}$ has full support on $\mathbb{R}$ and is stated formally in Assumption A.2. Since $E[g_h(\varepsilon_{1t})] = E[y_{t+h}]$ by iterated expectations, substituting into the ARF definition yields the density-ratio representation:

$$\mathrm{ARF}_h(\delta) \;=\; E\big[y_{t+h} \cdot (r_\delta(\varepsilon_{1t}) - 1)\big]. \tag{R}$$

Because $f$ is a univariate object, density-ratio estimation is a lower-dimensional nuisance problem than estimation of the conditional mean $g_h$.

The two representations place their estimation burden on different objects: the conditional mean $g_h$ and the shock density $f$, respectively. Having two paths to the same

estimand is what makes doubly robust estimation possible.

**Proposition 2.1** (Dual identification of the ARF)**.** *Suppose the potential-outcome representation in* (4) *holds, Assumption A.1 implies* $\varepsilon_{1t} \perp\!\!\!\perp U_{t+h}$*, and the overlap and integrability conditions in Assumptions A.2–A.3 hold for the fixed shift* $\delta$*. Then*

$$\mathrm{ARF}_h(\delta) = E[g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t})] = E[y_{t+h}\{r_\delta(\varepsilon_{1t}) - 1\}]. \tag{DI}$$

*Proof.* Independence of $\varepsilon_{1t}$ and $U_{t+h}$ gives $E[y_{t+h}(e)] = E[m_h(e, U_{t+h})] = g_h(e)$ for each shock value $e$ in the support. Substitution into (4) gives the conditional-mean representation (CM). For the density-ratio representation, the overlap condition permits the change of variables in (6), so $E[g_h(\varepsilon_{1t} + \delta)] = E[g_h(\varepsilon_{1t}) r_\delta(\varepsilon_{1t})]$. Since $E[g_h(\varepsilon_{1t})] = E[y_{t+h}]$, subtracting $E[y_{t+h}]$ gives (R). □

# 3 The Doubly Robust Estimator

The preceding section established that the average response function admits two observable representations: one through the conditional mean of the outcome given the shock, the other through the marginal density of the shock. Each suggests a plug-in estimator, and each carries a distinct vulnerability: the regression estimator inherits smoothing or approximation bias in the estimated conditional mean, while the reweighting estimator avoids the conditional mean but applies the density-ratio weight to the full outcome level, making it noisy when the ratio is variable in the tails. This section combines the two into a single estimator that uses the regression contrast as the primary estimate and applies the density-ratio weight only to the regression residual. The combination is doubly robust: first-stage errors affect the estimate only through their product, so consistency requires only that one of the two nuisance functions be consistently estimated. Deploying this estimator in a time-series setting requires three further elements: block cross-fitting to separate nuisance training from score evaluation, a product-rate condition under which the product bias is asymptotically negligible, and HAC inference targeting the long-run variance of the serially dependent influence score.

## 3.1 Construction of the DR Estimator

The two identification formulas imply two plug-in estimators. The conditional-mean representation gives

$$\widehat{\mathrm{ARF}}_h^{reg}(\delta) = \frac{1}{n_h}\sum_{t=1}^{n_h}\{\hat{g}_h(\varepsilon_{1t}+\delta) - \hat{g}_h(\varepsilon_{1t})\}, \tag{8}$$

while the density-ratio representation gives the pure reweighting estimator

$$\widehat{\mathrm{ARF}}_h^{rw}(\delta) = \frac{1}{n_h}\sum_{t=1}^{n_h} y_{t+h}\{\hat{r}_\delta(\varepsilon_{1t}) - 1\}. \tag{9}$$

The regression estimator is direct but transmits any error in $\hat{g}_h$ to the estimated response. The reweighting estimator avoids estimating $g_h$ altogether, but it weights the full outcome level and can therefore be noisy when the density ratio is variable. The doubly robust estimator combines both: it uses the regression contrast as the leading term and applies the density-ratio weight only to the regression residual:[1]

$$\widehat{\mathrm{ARF}}_h^{DR}(\delta) = \frac{1}{T-h}\sum_{t=1}^{T-h}\Big[\hat{g}_h(\varepsilon_{1t}+\delta) - \hat{g}_h(\varepsilon_{1t}) + (\hat{r}_\delta(\varepsilon_{1t}) - 1)(y_{t+h} - \hat{g}_h(\varepsilon_{1t}))\Big]. \tag{DR}$$

The first term, $\hat{g}_h(\varepsilon_{1t}+\delta) - \hat{g}_h(\varepsilon_{1t})$, is the regression term. The second term, $(\hat{r}_\delta(\varepsilon_{1t}) - 1)(y_{t+h} - \hat{g}_h(\varepsilon_{1t}))$, is the density-ratio bias correction applied to the regression residual. In the treatment effects literature, this construction is known as an augmented inverse-propensity-weighted (AIPW) estimator (Robins et al., 1994).

The doubly robust property of the estimator can be seen directly from the estimator formula. There are two cases to consider, corresponding to the two arms of the estimator.

*Regression arm.* Suppose first that the conditional-mean function is correctly specified, $\hat{g}_h = g_h$, but the density ratio may be misspecified. In this case the regression residual $y_{t+h} - g_h(\varepsilon_{1t})$ has conditional mean zero given $\varepsilon_{1t}$, by the definition of $g_h$ as the conditional expectation. The density-ratio correction term $(\hat{r}_\delta(\varepsilon_{1t}) - 1)(y_{t+h} - g_h(\varepsilon_{1t}))$ therefore also has conditional mean zero, regardless of whether $\hat{r}_\delta$ is a good estimate of $r_\delta$ or not. The estimator reduces to the regression representation plus mean-zero noise, and

[1] The cross-fitting fold index is suppressed to keep the score formula readable. The estimator used throughout the theory is the block cross-fitted version defined in Section 3.4: the nuisance functions $\hat{g}_h$ and $\hat{r}_\delta$ are estimated on temporally separated training observations, and each observation's score is evaluated using nuisance estimates that did not see that observation or its temporal neighbors. Thus (DR) should be read as defining the per-observation score, not as a recommendation to estimate and evaluate on the same sample.

consistency follows.

*Density-ratio arm.* Now suppose the density ratio is correctly specified, $\hat{r}_\delta = r_\delta$, but the conditional-mean estimate $\hat{g}_h$ may be wrong. This case is less immediate, because the correction term need not have conditional mean zero. Conditioning on $\varepsilon_{1t} = e$,

$$E[(r_\delta(e) - 1)\{y_{t+h} - \hat{g}_h(e)\} \mid \varepsilon_{1t} = e] = (r_\delta(e) - 1)\{g_h(e) - \hat{g}_h(e)\},$$

which is generally nonzero whenever $\hat{g}_h \neq g_h$. At first glance, this appears to introduce bias. The reason it does not is that, after averaging over the distribution of the shock, the bias contributed by the correction term exactly offsets the bias in the regression term. To see why, note that the density-ratio identity established in Section 2.3 implies that for any square-integrable function $\bar{g}$,

$$E[\bar{g}(E + \delta) - \bar{g}(E)] = E[(r_\delta(E) - 1)\bar{g}(E)],$$

where $E$ denotes a generic draw from the shock distribution. The left-hand side is the population regression term evaluated at $\bar{g}$; the right-hand side is the population reweighting term evaluated at $\bar{g}$. They are always equal when the density ratio is correct, regardless of whether $\bar{g}$ equals the true conditional mean. Combining the regression and correction terms in the full estimand gives

$$E[\bar{g}(E + \delta) - \bar{g}(E) + (r_\delta(E) - 1)\{Y - \bar{g}(E)\}] = \mathrm{ARF}_h(\delta),$$

so the estimator is consistent for any choice of $\bar{g}$, provided the density ratio is correct. As a limiting case, setting $\hat{g}_h = 0$ recovers the pure reweighting estimator $n_h^{-1} \sum_t y_{t+h}(\hat{r}_\delta(\varepsilon_{1t}) - 1)$.

The preceding argument established consistency by considering each arm in isolation. The following lemma shows a stronger property: the moment condition underlying the estimator is locally insensitive to small perturbations of either nuisance function around its true value. This insensitivity is known as Neyman orthogonality, and it is what allows the estimator to remain $\sqrt{n}$-consistent even when the nuisance functions are estimated at slower nonparametric rates. The reason is that the first-order effect of nuisance estimation error on the moment condition is exactly zero, so only the product of the two estimation errors contributes to the bias.

**Lemma 3.1** (Neyman orthogonality of the DR score)**.** *Let $W = (E, Y)$ denote a generic draw*

*from the bivariate marginal law of* $(\varepsilon_{1t}, y_{t+h})$*, let* $\theta_0 = \mathrm{ARF}_h(\delta)$*, and define*

$$\varphi(W;\theta,g,r) = g(E+\delta) - g(E) + \{r(E)-1\}\{Y - g(E)\} - \theta.$$

*At* $(\theta_0, g_h, r_\delta)$*, the moment is Neyman orthogonal to first-order perturbations of each nuisance. For every square-integrable perturbation* $a$ *of* $g_h$ *and* $b$ *of* $r_\delta$*,*

$$\frac{d}{d\tau}E[\varphi(W;\theta_0, g_h + \tau a, r_\delta)]\Big|_{\tau=0} = 0, \qquad \frac{d}{d\tau}E[\varphi(W;\theta_0, g_h, r_\delta + \tau b)]\Big|_{\tau=0} = 0.$$

*Proof.* For a perturbation $a$ of the regression nuisance,

$$\begin{aligned}\frac{d}{d\tau}E[\varphi(W;\theta_0, g_h + \tau a, r_\delta)]\Big|_{\tau=0} &= E[a(E+\delta) - a(E) - (r_\delta(E)-1)a(E)] \\ &= E[a(E+\delta)] - E[r_\delta(E)a(E)] \\ &= 0,\end{aligned}$$

where the last equality uses the density-ratio identity. For a perturbation $b$ of the ratio nuisance,

$$\frac{d}{d\tau}E[\varphi(W;\theta_0, g_h, r_\delta + \tau b)]\Big|_{\tau=0} = E[b(E)\{Y - g_h(E)\}] = E[b(E)E[Y - g_h(E) \mid E]] = 0.$$

□

## 3.2 Double Robustness

Neyman orthogonality is a local property: it concerns infinitesimal perturbations around the true nuisance functions. The double robustness result below shows that the same product structure delivers a global guarantee, in which one of the two nuisance components may be entirely misspecified.

**Proposition 3.1** (Double robustness)**.** *The DR estimator is consistent for* $ARF_h(\delta)$ *if either:*

*(a) the regression estimates* $\hat{g}_{k,h}$ *converge in probability to* $g_h$ *over the relevant domain, or*

*(b) the density-ratio estimates* $\hat{r}_{k,\delta}$ *converge in probability to* $r_\delta$*,*

*but not necessarily both, provided the misspecified component remains* $O_p(1)$ *in the corresponding* $L_2$ *norm. The formal statement, which specifies the block cross-fitted estimator of Section 3.4 and the uniformity and envelope conditions across folds, is given in Appendix A.*

*Sketch.* Let $k_h(t)$ denote the validation block containing observation $t$ and suppress the horizon subscript. The block cross-fitted estimator satisfies the expansion Writing $\Delta r_{k,t} = \hat{r}_{k_h(t),\delta}(\varepsilon_{1t}) - r_\delta(\varepsilon_{1t})$ and $\Delta g_{k,t} = \hat{g}_{k_h(t),h}(\varepsilon_{1t}) - g_h(\varepsilon_{1t})$ for the fold-specific estimation errors,

$$\widehat{\mathrm{ARF}}_h^{BCF} - \mathrm{ARF}_h = \underbrace{\frac{1}{n_h}\sum_{t=1}^{n_h} \psi_t^*}_{O_p(n_h^{-1/2})} - \underbrace{\frac{1}{n_h}\sum_{t=1}^{n_h} E\left[\Delta r_{k,t}\,\Delta g_{k,t} \,\middle|\, \mathcal{F}_{I_{k_h(t),h}}\right]}_{\text{fold-specific product bias}} + o_p(1), \tag{10}$$

with the remainder $o_p(n_h^{-1/2})$ under the stronger rate conditions used for inference. The key is the product structure of the bias: conditional on the training sample for each fold, the first-order derivative of the score with respect to each nuisance separately is zero, so only the product of the regression and density-ratio errors remains. If either error is zero, the product vanishes regardless of the other. This is the double-robustness result specific to the conditional-mean and density-ratio nuisances in the nonparametric ARF problem.

The practical implication is that the estimator provides insurance against misspecification of either nuisance component. A researcher who is confident in the conditional-mean model but uncertain about the shock density still obtains a consistent estimate, and vice versa. This guarantee concerns consistency; the rate at which the product bias vanishes, and the additional conditions needed for valid inference, are the subject of the next subsection.

## 3.3 Asymptotic Normality and HAC Inference

For valid inference, both nuisance estimators must converge individually, and the product of their errors must vanish fast enough to be asymptotically negligible. The following conditions are sufficient:

$$\begin{aligned}
\max_{1\le k\le K}\{\|\hat{g}_{k,h} - g_h\|_{2,f} + \|\hat{g}_{k,h} - g_h\|_{2,f_\delta}\} &= o_p(1),\\
\max_{1\le k\le K}\{\|\hat{r}_{k,\delta} - r_\delta\|_{2,f} + \|\hat{r}_{k,\delta} - r_\delta\|_{2,\sigma}\} &= o_p(1),\\
\max_{1\le k\le K}\|\hat{g}_{k,h} - g_h\|_{2,f}\cdot\|\hat{r}_{k,\delta} - r_\delta\|_{2,f} &= o_p(n_h^{-1/2}),
\end{aligned} \tag{11}$$

where $\|\cdot\|_{2,f}$ denotes the $L_2$ norm under the shock distribution ($\|a\|_{2,f}^2 = E[a(\varepsilon_{1t})^2]$), $\|\cdot\|_{2,f_\delta}$ the $L_2$ norm under the shifted distribution ($\|a\|_{2,f_\delta}^2 = E[a(\varepsilon_{1t}+\delta)^2]$), and $\|\cdot\|_{2,\sigma}$ the conditional-variance-weighted norm ($\|a\|_{2,\sigma}^2 = E[a(\varepsilon_{1t})^2\sigma_h^2(\varepsilon_{1t})]$, with $\sigma_h^2(e) = \mathrm{Var}(y_{t+h} \mid \varepsilon_{1t} = e)$). The first two lines require each nuisance to be individually

consistent in the norms that appear in the estimator's moment condition. The unweighted $\|\cdot\|_{2,f}$ norm for the density ratio enters through the product-bias term, while the $\sigma$-weighted norm controls residual terms of the form $(\hat{r}_{k,\delta} - r_\delta)(y_{t+h} - g_h(\varepsilon_{1t}))$. The third line is the product-rate condition: it does not require either component to converge at $n_h^{-1/4}$ individually, only that the product of their rates beats $n_h^{-1/2}$.

Under these conditions and the block cross-fitting and weak-dependence regularity stated in Appendix A.1, the DR estimator admits the linear expansion

$$\sqrt{n_h}\big(\widehat{\mathrm{ARF}}_h^{BCF}(\delta) - \mathrm{ARF}_h(\delta)\big) = \frac{1}{\sqrt{n_h}} \sum_{t=1}^{n_h} \psi_t^* + o_p(1), \tag{12}$$

where $\psi_t^*$ is the efficient influence function recorded in the following proposition; the derivation and a variance decomposition are deferred to Appendix A.3 and Appendix A.4.

**Proposition 3.2** (Influence score)**.** *Under regularity conditions stated in Appendix A, the efficient influence function for* $\mathrm{ARF}_h(\delta)$ *in the unrestricted bivariate marginal model is*

$$\psi_t^* = \big[g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t})\big] + \big[r_\delta(\varepsilon_{1t}) - 1\big]\big[y_{t+h} - g_h(\varepsilon_{1t})\big] - \mathrm{ARF}_h(\delta). \tag{EIF}$$

The linear expansion therefore gives

$$\sqrt{n_h}\big(\widehat{\mathrm{ARF}}_h^{BCF}(\delta) - \mathrm{ARF}_h(\delta)\big) \xrightarrow{d} N(0, \Sigma_h^*), \tag{13}$$

where the asymptotic variance is the long-run variance of the influence-function process:

$$\Sigma_h^* = \sum_{j=-\infty}^{\infty} \mathrm{Cov}(\psi_t^*, \psi_{t-j}^*). \tag{LRV}$$

This equals $\mathrm{Var}(\psi_t^*)$ only when all nonzero-lag autocovariances vanish. A dynamic structural recursion can generate serial dependence in $\psi_t^*$ even at $h = 0$, because $y_t$ may depend on lagged states. Inference therefore targets $\Sigma_h^*$ using a HAC estimator under stationary weak dependence.

Because the structural time-series model may impose restrictions beyond the bivariate marginal law, the contemporaneous variance $\mathrm{Var}(\psi_t^*)$ should be interpreted as the efficiency bound for the enlarged marginal model unless equivalence of the structural and marginal tangent spaces is separately established. This qualification is not needed for HAC inference: Proposition A.2 uses $\psi_t^*$ as the first-order score and estimates its long-run variance.

The influence score also clarifies what the DR correction does and does not provide relative to a pure regression approach. A regression plug-in estimator that is undersmoothed, bias-corrected, or otherwise tuned so that the ARF functional is root-$T$ regular has the same first-order influence score $\psi_t^*$, so the DR correction is not a generic first-order variance improvement. Its practical role is bias protection: with conventional smoothing or misspecified function classes, the DR bias is governed by a product of regression and density-ratio errors, whereas a regression-only HAC interval remains centered at a potentially biased plug-in estimate. This distinction matters for coverage. A HAC estimator targets the long-run variance of the score sequence; it does not remove deterministic smoothing or approximation bias in the center of a regression-only interval. Thus a regression-only estimator can have lower RMSE but worse coverage if its bias is small in levels yet nonnegligible relative to its standard error. The Monte Carlo coverage panels in Section 4 evaluate this implication directly.

The rate conditions above are stated in terms of $L_2$ norms without specifying which estimators achieve them. Appendix B verifies these conditions for one concrete choice: kernel estimators of the conditional mean and the shock density, with bandwidths satisfying standard nonparametric rate requirements. Other first-stage estimators, such as sieves or local polynomials, could be substituted provided they deliver the same product rate.

### 3.4 Block Cross-Fitting

In an i.i.d. setting, cross-fitting randomly partitions the sample into folds, estimates nuisance functions on each fold's complement, and evaluates the score on the held-out fold. Random partitioning ensures independence between the training and evaluation samples. With time-series data, random assignment would break the temporal ordering, and adjacent observations are dependent, so two modifications are needed: the folds must be contiguous blocks that preserve the time structure, and a buffer of discarded observations must separate each evaluation block from its training sample to ensure approximate independence.

For fixed horizon $h$, set $n_h = T - h$ and $\mathcal{T}_h = \{1, \ldots, n_h\}$. Partition $\mathcal{T}_h$ into a fixed number $K$ of non-overlapping contiguous validation blocks $B_{1,h}, \ldots, B_{K,h}$ with lengths proportional to $n_h$. For a block $B_{k,h} = [a_{k,h}, b_{k,h}]$, choose a buffer length $m_T$ satisfying $m_T \to \infty$ and $m_T/n_h \to 0$, and define

$$C_{k,h}(m_T) = \{s \in \mathcal{T}_h : a_{k,h} - m_T \leq s \leq b_{k,h} + m_T\}, \tag{14}$$

$$I_{k,h} = \mathcal{T}_h \setminus C_{k,h}(m_T). \tag{15}$$

The nuisance functions $\hat{g}_{k,h}$ and $\hat{r}_{k,\delta}$ are estimated only from observations indexed by $I_{k,h}$. For $t \in B_{k,h}$, define

$$\hat{\phi}_{t,h}^{BCF} = \hat{g}_{k,h}(\varepsilon_{1t} + \delta) - \hat{g}_{k,h}(\varepsilon_{1t}) + \{\hat{r}_{k,\delta}(\varepsilon_{1t}) - 1\}\{y_{t+h} - \hat{g}_{k,h}(\varepsilon_{1t})\}. \tag{16}$$

The estimator is

$$\widehat{\mathrm{ARF}}_h^{BCF}(\delta) = \frac{1}{n_h} \sum_{k=1}^{K} \sum_{t \in B_{k,h}} \hat{\phi}_{t,h}^{BCF}. \tag{17}$$

The formal conditions require the buffer length $m_T$ to diverge slowly enough that the training sample retains a fixed fraction of the observations, while the mixing coefficients of the process decay fast enough at the buffer distance to deliver approximate independence. Algorithm 1 summarizes the complete procedure, including the outer loop over horizons and the HAC standard error step.

**Algorithm 1** Block Cross-Fitted Doubly Robust Estimator of $\mathrm{ARF}_h(\delta)$

**Require:** $\{y_t, \varepsilon_{1t} : t = 1, \ldots, T\}$, shock size $\delta$, maximum horizon $H$, number of folds $K$, buffer length $m_T$, and fold-specific estimators for $g_h$ and $r_\delta$.

```
for h = 0, ..., H do
    Set n_h = T − h and partition {1, ..., n_h} into contiguous validation blocks B_{1,h}, ..., B_{K,h}.
    for k = 1, ..., K do
        Form the buffer-excluded training set I_{k,h}.
        Estimate ĝ_{k,h} from {(ε_{1s}, y_{s+h}) : s ∈ I_{k,h}} and estimate r̂_{k,δ} from {ε_{1s} : s ∈ I_{k,h}}.
        Evaluate φ̂^{BCF}_{t,h} in (16) for all t ∈ B_{k,h}.
    end for
    Average the ordered held-out scores to obtain ÂRF^{BCF}_h(δ) in (17).
    Estimate standard errors from a HAC estimator applied to {φ̂^{BCF}_{t,h}}_{t=1}^{n_h}.
end for
```

**Ensure:** $\{\widehat{\mathrm{ARF}}_h^{BCF}(\delta) : h = 0, \ldots, H\}$.

The buffer length $m_T$ must be chosen to balance two competing requirements: it must grow fast enough to attenuate dependence between the training and evaluation samples, yet slowly enough that the training set retains a fixed fraction of the observations. A practical default that satisfies both is

$$m_T = \max\{h, \lceil n_h^{1/4} \rceil\},$$

which ensures $m_T \to \infty$ and $m_T / n_h \to 0$ while keeping the buffer at least as wide as the

forecast horizon. Additional implementation choices, including common-horizon fold geometry and HAC bandwidth selection, are collected in Appendix C.

### 3.5 Density Ratio Estimation

The estimator in (DR) requires an estimate $\hat{r}_\delta$ of the one-dimensional density ratio $r_\delta(e) = f(e-\delta)/f(e)$. The default approach is nonparametric: estimate the marginal shock density $f$ on each buffer-excluded training fold and form the plug-in ratio

$$\hat{r}_{k,\delta}(e) = \frac{\hat{f}_k(e-\delta)}{\hat{f}_k(e)}. \tag{18}$$

When the marginal shock distribution is credibly parametric, whether because a distributional assumption is maintained as part of the identification normalization or because the simulation design fixes it, a parametric plug-in ratio may be used instead. In either case, the density-ratio estimate for each fold is trained on the same buffer-excluded observations as the outcome regression, so each evaluated score remains held out from both nuisance estimators.

Large shift sizes $\delta$ create a tail-instability problem. The density ratio compares the shock density at two points separated by $\delta$, so when $\delta$ is large relative to the spread of the distribution the denominator $\hat{f}_k(e)$ can be near zero in the tails, making the estimated ratio volatile. The difficulty is analogous to the overlap problem in treatment-effects estimation, where inverse-propensity weights become unstable as the propensity score approaches zero or one. Appendix C describes the tail regularization and ratio caps used to control this instability; Remark B.2 discusses the interpretation of the resulting stabilized estimand.

Taken together, the estimator combines three elements. The first is the doubly robust score (DR), which uses the regression contrast as the primary estimate and applies the density-ratio weight only to the residual. The second is block cross-fitting with buffer separation, which ensures that nuisance estimation and score evaluation are approximately independent despite serial dependence. The third is HAC inference targeting the long-run variance of the influence score. The double-robustness property guarantees consistency when either nuisance component is misspecified. The product-rate condition ensures that the joint estimation error is asymptotically negligible for inference.

# 4 Monte Carlo Simulations

Two sets of simulations evaluate the doubly robust estimator. The first demonstrates the formal double-robustness property: when the regression arm is structurally misspecified, the density-ratio correction eliminates the asymptotic bias that a standalone regression estimator carries. The second examines the finite-sample cost of the correction when the regression is already well matched to the data-generating process (DGP). Because the DR estimator and a well-tuned regression plug-in share the same first-order influence score (Proposition 3.2), the RMSE gap in this second exercise reflects finite-sample variation from the residual correction rather than a first-order efficiency loss. For each design, the figures report both point-estimation behavior and inference: response estimates, mean bias, RMSE, and empirical coverage of nominal 90% HAC confidence intervals. The coverage panels are central because the DR correction is aimed at reducing centering error from smoothing or misspecification bias, not at delivering a generic variance reduction. All figures are collected in Appendix D.

## 4.1 Design

The DGP is adapted from Gonçalves et al. (2024a). The shock is $x_t = \varepsilon_{1t}$, where $\varepsilon_{1t}$ and $\varepsilon_{2t}$ are independent standard normal draws. The outcome follows

$$y_t = 0.5\, y_{t-1} + 0.5\, x_t + 0.3\, x_{t-1} - 0.4\, q(x_t) - 0.3\, q(x_{t-1}) + \varepsilon_{2t}, \tag{19}$$

and the estimand is the unconditional average response function (4). I consider two functional forms for $q$ that pose complementary challenges for nonparametric estimation:

- *ReLU:* $q(x) = \max(x, 0)$. The kink at the origin is poorly approximated by smooth polynomial bases, so the power series estimator exhibits persistent finite-sample bias. The local linear estimator, which imposes no global smoothness, adapts well.
- *Cubic:* $q(x) = x^3$. The high curvature induces substantial bandwidth-driven bias in the local linear estimator. The power series estimator, which nests the true specification, performs well.

The ReLU design is a deliberate nonsmooth stress test; it is not covered by the $C^2$ primitive rate verification in Proposition B.1. Each replication discards $B = 1{,}000$ burn-in observations before recording a sample of length $T$.

The simulation varies $T \in \{250, 500, 1{,}000, 2{,}000\}$ and $\delta \in \{1, 2\}$, with $R = 5{,}000$ replications per design cell. Results are presented as four-row multi-panel figures, one

for each combination of functional form and shock size: columns index the sample sizes shown in the figure, and rows display the estimated impulse response (median and interquartile range), mean bias, root mean squared error, and empirical coverage of nominal 90% HAC confidence intervals.

Each figure compares two estimators: a standalone regression estimator of $g_h$ and the block cross-fitted DR estimator (DR), which pairs the same regression $\hat{g}_h$ with the fold-specific parametric Gaussian density ratio

$$\hat{r}_{k,\delta}(e) = \exp\Big(\frac{\delta e}{\hat{\sigma}^2_{1,k}} - \frac{\delta^2}{2\hat{\sigma}^2_{1,k}}\Big), \tag{20}$$

where $\hat{\sigma}^2_{1,k}$ is estimated from the same buffer-excluded training shocks used for the other fold-$k$ nuisance estimates. Because the shock is Gaussian by construction, this density-ratio model is correctly specified. Using the parametric ratio throughout both sets of simulations isolates the contribution of the bias-correction term from the separate question of how to estimate the density ratio nonparametrically.

The regression specification varies across the two exercises and is described in each subsection. The DR implementation uses Algorithm 1 with contiguous folds and horizon-aware buffers.

## 4.2 Double robustness under structural misspecification

The first set of simulations illustrates the double-robustness property of Proposition 3.1 in finite samples. The regression arm is a linear local projection that omits the nonlinear term entirely,

$$y_{t+h} = \alpha_h + \psi_h \, \varepsilon_{1t} + \omega_{t+h},$$

estimated by OLS. This is the specification a researcher would adopt if unaware of the nonlinearity. It produces a bias that does not vanish with $T$: the linear projection $\psi_h$ captures only the linear component of the ARF, missing the contribution of $q$. The DR estimator pairs this misspecified regression with the parametric density ratio (20). The linear LP is estimated by OLS of $y_{t+h}$ on a constant and $\varepsilon_{1t}$.

Figures 1–4 report the results for both functional forms. The standalone linear LP (dashed) exhibits bias that persists across all sample sizes—the hallmark of structural misspecification rather than finite-sample imprecision. The DR estimator (solid) eliminates the vast majority of this bias. Because the density ratio is correctly specified, the augmentation term $(r_\delta - 1)(y_{t+h} - \hat{g}_h)$ reweights the regression residuals to recover

the nonlinear signal that the linear projection discards. The bias reduction is most pronounced for $\delta = 2$, where the nonlinear component of the ARF is largest.

The coverage panels give the inferential counterpart to the bias panels. The regression-only HAC intervals severely undercover across the misspecified designs, as expected when intervals are centered at a structurally biased estimator. HAC standard errors account for sampling variation around the estimated score, but they do not remove the omitted-nonlinearity bias in the center of the interval. The DR intervals have substantially higher coverage because the correctly specified density ratio carries the residual correction. In the ReLU designs and in the cubic design with $\delta = 1$, coverage is close to the nominal 90% level over most horizons and sample sizes. In the more severe cubic design with $\delta = 2$, DR coverage still falls below the nominal target at short and intermediate horizons, reflecting the finite-sample difficulty created by large shifts and high curvature, but it remains far above the regression-only benchmark.

### 4.3 Finite-sample trade-off with flexibly estimated regression

The second set of simulations examines the finite-sample trade-off when the robustness correction is not needed for consistency under the intended flexible regression specification. The regression arm is now the well-matched nonparametric method: local linear for the ReLU DGP and power series for the cubic DGP. The local linear estimator uses a Gaussian kernel with a Fan and Gijbels (1996) rule-of-thumb bandwidth based on a preliminary polynomial of order 2; the power series estimator uses polynomial order $0.5\,T^{1/3}$ rounded to the nearest integer. Both follow Gonçalves et al. (2024a).

Figures 5–8 show a different trade-off from the misspecification exercise. The standalone regression estimator generally has low bias and low RMSE, especially in the smaller-shift designs. This is the case in which the DR correction has little first-order bias to remove. The residual correction therefore mainly adds density-ratio-weighted residual variation, and the resulting RMSE cost can be substantial when the shock shift is large. This is most visible for $\delta = 2$, where the Gaussian density ratio is more variable and residual weighting amplifies finite-sample noise. The RMSE gap should therefore be interpreted as a finite-sample AMSE trade-off induced by the particular smoothing and density-ratio choices, not as a direct estimate of a first-order efficiency loss relative to a root-$T$ regular regression functional.

The coverage panels show why RMSE alone does not summarize the inferential trade-off. Even in the flexible-regression designs, regression-only HAC intervals generally undercover. The reason is that their pointwise smoothing or approximation bias, while

often small on the scale of the response itself, can be nonnegligible relative to the estimated standard error. HAC standard errors estimate stochastic long-run variation; they do not recenter intervals for deterministic smoothing bias. The DR intervals have substantially higher coverage in most flexible-regression cells because the orthogonal score reduces this centering error and because the additional residual-weight variation is reflected in the HAC standard errors. The improvement should not be read as uniform nominal coverage: the large-shift ReLU and cubic designs still display finite-sample undercoverage. The appropriate conclusion is that the DR correction materially improves coverage at the same smoothing choices, while leaving a visible finite-sample variance cost in difficult large-shift designs.

Between the two polar cases lies a middle ground of correct but slow specification, where the function class contains the true $g_h$ but the estimator converges slowly due to bandwidth choice, smoothness, or finite-sample approximation. The product-of-errors property (Proposition 3.1) implies that the DR correction provides insurance in this intermediate regime as well: even when both nuisance components contain estimation error, the bias of the DR estimator is proportional to their product, so a well-specified density ratio can attenuate the finite-sample bias of a slow-converging regression. A regression-only procedure would need undersmoothing or explicit bias correction to target comparable coverage at the same nominal level.

Taken together, the two sets of simulations support the use of the DR estimator when regression misspecification or smoothing bias is a serious concern. Under structural misspecification, the correction delivers large bias reductions and much higher coverage. With flexible regression arms, the correction can raise RMSE, sometimes substantially for large shifts, but the coverage panels show that this variance cost buys better-calibrated inference than regression-only HAC intervals at the same smoothing choices.

# 5 Conclusion

Nonparametric local projections allow researchers to trace impulse responses without restricting how shocks propagate. Regression-only implementations, however, stake estimation and inference on a single nonparametric conditional mean for each horizon. This paper shows that the same average response function admits a second, density-ratio representation. Combining the two yields a doubly robust estimator: a regression estimate augmented with a density-ratio-weighted residual correction. Consistency requires only that either the conditional mean or the density ratio be consistently estimated. First-stage errors enter only through their product, and buffer-separated block

cross-fitting renders this product bias asymptotically negligible under a product-rate condition.

The influence-function analysis clarifies the role of the augmentation term. A regression plug-in estimator that is already root-$T$ regular shares the same first-order influence score, so the DR correction does not provide a generic variance improvement. Its value is bias protection and inferential centering. When the regression arm is misspecified or biased by smoothing, the density-ratio correction removes the leading bias component. When the regression is already accurate, the same correction adds residual-weight variation without improving first-order efficiency.

The Monte Carlo simulations confirm this inferential trade-off. Under structural misspecification, the DR correction sharply reduces bias and raises coverage relative to regression-only HAC intervals. With flexible regression arms, the correction often raises RMSE, sometimes substantially for large shifts, but it still improves coverage because the orthogonal score reduces the smoothing-bias component that HAC standard errors do not remove. The evidence therefore supports using the DR estimator when the main concern is reliable inference under possible regression bias, while recognizing the finite-sample variance cost of residual weighting.

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# A Proofs of Main Results

This appendix section collects the high-level regularity conditions, formal statements, and proofs of the main results. The influence-function derivation, one sufficient kernel-rate verification, and implementation details are separated below.

## A.1 Regularity Conditions

**Assumption A.1.** $\varepsilon_t = (\varepsilon_{1t}, \ldots, \varepsilon_{nt})'$ is i.i.d. across $t$ with mean zero, and the components $\varepsilon_{1t}, \ldots, \varepsilon_{nt}$ are mutually independent (with variances $\sigma_i^2$).

**Assumption A.2.** The density $f$ of $\varepsilon_{1t}$ is bounded, bounded away from zero on compact sets, and satisfies $E[r_\delta(\varepsilon_{1t})^2] < \infty$.

**Assumption A.3.** $E[y_{t+h}^2] < \infty$ and $E[y_{t+h}^2 \cdot r_\delta(\varepsilon_{1t})^2] < \infty$.

**Assumption A.4** (Stationarity and weak dependence)**.** For each fixed horizon $h$, the observed process $W_{t,h} = (\varepsilon_{1t}, y_{t+h})$ is strictly stationary and strong mixing with coefficients $\alpha(j)$. There exists $\eta > 0$ such that $E|\psi_t^*|^{2+\eta} < \infty$, $\sum_{j=1}^{\infty} \alpha(j)^{\eta/(2+\eta)} < \infty$, and the central limit theorem and HAC consistency conditions hold for the influence-function process $\{\psi_t^*\}$. The same mixing and moment bounds apply to the fold-specific nuisance-residual products appearing in Lemma A.1.

**Assumption A.5** (Fold-specific nuisance rates)**.** Let $n_h = T - h$, $\|a\|_{2,f}^2 = E[a(\varepsilon_{1t})^2]$, $\|a\|_{2,f_\delta}^2 = E[a(\varepsilon_{1t} + \delta)^2]$, and $\|b\|_{2,\sigma}^2 = E[b(\varepsilon_{1t})^2 \sigma_h^2(\varepsilon_{1t})]$, where $\sigma_h^2(e) = \mathrm{Var}(y_{t+h} \mid \varepsilon_{1t} = e)$. For each fold $k$, let $\hat{g}_{k,h}$ and $\hat{r}_{k,\delta}$ denote the nuisance estimators trained on $I_{k,h}$. The following hold uniformly over the fixed number of folds:

(a) For consistency: either

$$\max_k \{\|\hat{g}_{k,h} - g_h\|_{2,f} + \|\hat{g}_{k,h} - g_h\|_{2,f_\delta}\} = o_p(1)$$

or

$$\max_k \{\|\hat{r}_{k,\delta} - r_\delta\|_{2,f} + \|\hat{r}_{k,\delta} - r_\delta\|_{2,\sigma}\} = o_p(1),$$

and the misspecified component is $O_p(1)$ in the corresponding unweighted or residual-weighted norm. Trimmed and capped nuisance estimators can enforce finite-sample envelopes, but if the caps diverge with the fold training size the boundedness condition is imposed rather than automatic;

(b) For asymptotic linearity:

$$\max_k \{\|\hat{g}_{k,h} - g_h\|_{2,f} + \|\hat{g}_{k,h} - g_h\|_{2,f_\delta}\} = o_p(1),$$

$$\max_k \{\|\hat{r}_{k,\delta} - r_\delta\|_{2,f} + \|\hat{r}_{k,\delta} - r_\delta\|_{2,\sigma}\} = o_p(1),$$

and

$$\max_k \|\hat{g}_{k,h} - g_h\|_{2,f} \cdot \|\hat{r}_{k,\delta} - r_\delta\|_{2,f} = o_p(n_h^{-1/2}).$$

**Assumption A.6** (Sequential block cross-fitting)**.** For each fixed horizon $h$, the index set $\mathcal{T}_h = \{1, \ldots, n_h\}$ is partitioned into a fixed number $K$ of contiguous validation blocks $B_{1,h}, \ldots, B_{K,h}$ such that $\min_k |B_{k,h}|/n_h$ and $\max_k |B_{k,h}|/n_h$ are bounded away from zero and infinity. For $B_{k,h} = [a_{k,h}, b_{k,h}]$, let

$$C_{k,h}(m_T) = \{s \in \mathcal{T}_h : a_{k,h} - m_T \leq s \leq b_{k,h} + m_T\}, \qquad I_{k,h} = \mathcal{T}_h \setminus C_{k,h}(m_T).$$

The nuisance estimators $\hat{g}_{k,h}$ and $\hat{r}_{k,\delta}$ are measurable with respect to the sigma-field generated by observations in $I_{k,h}$. The buffer satisfies $m_T \to \infty$, $m_T/n_h \to 0$, and $n_h^{1/2}\alpha(m_T)^{\eta/(2+\eta)} \to 0$, where $\alpha(\cdot)$ and $\eta$ are as in Assumption A.4. Consequently, the training sigma-field for fold $k$ is asymptotically separated from the held-out block $B_{k,h}$ at the rate needed for the block empirical-process bounds in Lemma A.1.

**Assumption A.7** (HAC bandwidth and estimated-score replacement)**.** Let

$$\phi^*_{t,h} = g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t}) + \{r_\delta(\varepsilon_{1t}) - 1\}\{y_{t+h} - g_h(\varepsilon_{1t})\}$$

denote the uncentered oracle score, so that $\psi^*_t = \phi^*_{t,h} - \mathrm{ARF}_h(\delta)$. Let $d_{t,h} = \hat{\phi}^{BCF}_{t,h} - \phi^*_{t,h}$. The HAC bandwidth $B_T$ satisfies $B_T \to \infty$, $B_T/n_h \to 0$, and

$$B_T \left( \frac{1}{n_h} \sum_{t=1}^{n_h} d^2_{t,h} \right)^{1/2} = o_p(1).$$

A sufficient nuisance-level condition is that, with $\Delta g_k = \hat{g}_{k,h} - g_h$ and $\Delta r_k = \hat{r}_{k,\delta} - r_\delta$, the score-level error

$$\max_k \left\{ \|\Delta g_k\|_{2,f_\delta} + \|r_\delta \Delta g_k\|_{2,f} + \|\Delta r_k\|_{2,\sigma} + \|\Delta r_k \Delta g_k\|_{2,f} \right\}$$

is $o_p(B_T^{-1})$, with the corresponding empirical norms controlled by the block cross-fitted validation blocks. For the kernel rates in Proposition B.1, this permits standard HAC bandwidths satisfying $B_T = o(n_h^{2/5-o(1)})$ when the displayed score-level nuisance bound has the same order as the population nuisance rates.

*Remark* A.1 (On Assumption A.2). The condition $E[r_\delta(\varepsilon_{1t})^2] < \infty$ is a direct overlap and tail requirement on the shifted density ratio. It should be verified for the shock distribution at the shift under study. Many regular polynomial-tailed location families have $r_\delta(e) \to 1$ in the tails for fixed $\delta$, so polynomial tails do not automatically violate the condition. By contrast, very light or irregular tails can make the ratio large on parts of the support. The boundedness of $r_\delta$ itself is not assumed.

*Remark* A.2 (On Assumption A.3). The condition $E[y^2_{t+h} r^2_\delta] < \infty$ is the key moment restriction. It is satisfied when both $y_{t+h}$ and $\varepsilon_{1t}$ have sub-exponential or lighter tails, but requires verification for specific DGPs. This condition plays the same role as the bounded support assumption in the treatment effects literature — it ensures the reweighted moments are well-defined, but is weaker.

## A.2 Block Cross-Fitted Empirical Remainders

The next lemma records the empirical-process expansion used in both consistency and asymptotic normality. It is stated at the high-level nuisance-rate layer; the primitive kernel rates used to verify those nuisance conditions are collected separately in Appendix B.

**Lemma A.1** (Block cross-fitted empirical remainders)**.** *Let* $\Delta g_k = \hat{g}_{k,h}^T - g_h$ *and* $\Delta r_k = \hat{r}_{k,\delta}^T - r_\delta$*. Under Assumptions A.4 and A.6,*

$$\widehat{\mathrm{ARF}}_h^{BCF} - \mathrm{ARF}_h = \frac{1}{n_h}\sum_{t=1}^{n_h}\psi_t^* - \sum_{k=1}^{K}\frac{|B_{k,h}|}{n_h}P\{\Delta r_k(\varepsilon_{1t})\Delta g_k(\varepsilon_{1t})\} + R_{n,h}, \tag{21}$$

*where* $R_{n,h} = o_p(1)$ *under Assumption A.5(a) and* $R_{n,h} = o_p(n_h^{-1/2})$ *under Assumption A.5(b). Consequently, the empirical product-bias term is controlled by the fold-specific population* $L_2(P_f)$ *rates in Proposition B.1.*

*Proof.* For $t \in B_{k,h}$, decompose

$$\begin{aligned}\phi(W_{t,h};\hat{\eta}_k) - \phi(W_{t,h};\eta_0) = P\{\phi(W;\hat{\eta}_k) - \phi(W;\eta_0)\} \\ + \{\phi(W_{t,h};\hat{\eta}_k) - \phi(W_{t,h};\eta_0) - P[\phi(W;\hat{\eta}_k) - \phi(W;\eta_0)]\}.\end{aligned}$$

The population term equals $-P\{\Delta r_k \Delta g_k\}$ by the density-ratio identity and conditional mean-zero property of $y_{t+h} - g_h(\varepsilon_{1t})$. For the centered term, condition on the training sigma-field generated by $I_{k,h}$. Because the validation block is separated from the training observations by $m_T$, the strong-mixing covariance inequality and the buffer condition in Assumption A.6 allow the conditional second moment of the validation-block average to be bounded, up to $o(n_h^{-1})$ terms, by a constant times

$$\frac{|B_{k,h}|}{n_h^2}\left(\|\Delta g_k\|_{2,f}^2 + \|\Delta g_k\|_{2,f_\delta}^2 + \|\Delta r_k\|_{2,\sigma}^2 + \|\Delta r_k \Delta g_k\|_{2,f}^2\right).$$

Summing over the fixed number of folds gives a centered empirical-process term of order

$$O_p\left(n_h^{-1/2}\max_k\{\|\Delta g_k\|_{2,f} + \|\Delta g_k\|_{2,f_\delta} + \|\Delta r_k\|_{2,\sigma} + \|\Delta r_k \Delta g_k\|_{2,f}\}\right) + o_p(n_h^{-1/2}).$$

This is $o_p(1)$ under the consistency part of Assumption A.5 and $o_p(n_h^{-1/2})$ under Assumption A.5(b). The product drift is bounded by Cauchy–Schwarz:

$$|P\{\Delta r_k(\varepsilon_{1t})\Delta g_k(\varepsilon_{1t})\}| \le \|\Delta r_k\|_{2,f}\,\|\Delta g_k\|_{2,f}.$$

This proves (21). □

*Remark* A.3 (Same-sample and leave-one-out variants). The block cross-fitted estimator is the default estimator in the paper because it separates nuisance training from score

evaluation. A same-sample kernel estimator can be asymptotically equivalent if one proves the relevant stochastic-equicontinuity or U-statistic projection remainder directly. Similarly, a leave-one-out regression residual removes the mechanical own-observation fit but does not separate the held-out score from neighboring observations in a dynamic time series and does not address own-observation fitting in the density-ratio nuisance. For this reason, leave-one-out fitting is best treated as a computational sensitivity check rather than as the formal route to the main theorem.

## A.3 Efficient Influence Function

I derive the efficient influence function for $\theta \equiv \mathrm{ARF}_h(\delta) = E[g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t})]$ stated in Proposition 3.2. The derivation proceeds in three steps: I characterize the tangent space of the semiparametric model, compute the pathwise derivative of $\theta$, and verify that $\psi_t^*$ is the Riesz representer of this derivative in the tangent space.

**Step 1. Tangent space.** The functional $\theta$ depends on the distribution of the full time series $\{z_t\}$ only through the bivariate marginal of $(\varepsilon_{1t}, y_{t+h})$. I work in the *bivariate marginal model*: the semiparametric model that treats the joint density of this pair, $p(e, y) = f(e)\, c(y \mid e)$, as the observed-data law, leaving both the marginal density $f$ and the conditional density $c(\cdot \mid e)$ unrestricted (subject to Assumptions A.2–A.3). See Remark A.4 below for the relationship between this model and the full structural model.

Consider a smooth one-dimensional submodel $\{P_\eta : \eta \in (-\epsilon, \epsilon)\}$ through the true distribution $P = P_0$, with score

$$s(e, y) = \frac{\partial}{\partial \eta} \log p_\eta(e, y)\Big|_{\eta=0} = s_1(e) + s_2(e, y), \tag{22}$$

where $s_1(e) = \frac{\partial}{\partial \eta} \log f_\eta(e)\big|_{\eta=0}$ is the marginal score and $s_2(e, y) = \frac{\partial}{\partial \eta} \log c_\eta(y \mid e)\big|_{\eta=0}$ is the conditional score. These satisfy $E[s_1(\varepsilon_{1t})] = 0$ and $E[s_2(\varepsilon_{1t}, y_{t+h}) \mid \varepsilon_{1t}] = 0$ almost surely.

The tangent space $\mathcal{T}$ of the bivariate marginal model is the set of all such scores:

$$\mathcal{T} = \big\{ s_1(\varepsilon_{1t}) + s_2(\varepsilon_{1t}, y_{t+h}) : s_1 \in L_2^0(f),\ s_2 \in L_2(p),\ E[s_2 \mid \varepsilon_{1t}] = 0 \big\}. \tag{23}$$

This is the full space $L_2^0(P)$: any mean-zero square-integrable function $h(e, y)$ can be decomposed as $h(e, y) = E[h(\varepsilon_{1t}, y_{t+h}) \mid \varepsilon_{1t} = e] + \big(h(e, y) - E[h(\varepsilon_{1t}, y_{t+h}) \mid \varepsilon_{1t} = e]\big)$, where the first term has mean zero (by iterated expectations) and the second has conditional mean zero given $\varepsilon_{1t}$ (Lemma A.2).

**Lemma A.2** (Tangent space of the bivariate marginal model)**.** *In the bivariate marginal model that treats $p(e, y) = f(e)\, c(y \mid e)$ as the observed-data density with both $f$ and $c$ unrestricted (subject to Assumptions A.2–A.3), the tangent space equals* (23)*, i.e. the full $L_2^0(P)$.*

*Proof.* In a nonparametric model with density $p(e, y) = f(e)\, c(y \mid e)$ where $f$ and $c$ are both unrestricted, the marginal and conditional components can be perturbed

independently. A marginal perturbation $f_\eta(e) = f(e)(1+\eta\, s_1(e))$ with $s_1 \in L_2^0(f)$ is a valid submodel for any such $s_1$, holding $c$ fixed. A conditional perturbation $c_\eta(y \mid e) = c(y \mid e)(1+\eta\, s_2(e,y))$ with $E[s_2 \mid \varepsilon_{1t} = e] = 0$ is a valid submodel for any such $s_2$, holding $f$ fixed. Every mean-zero $h \in L_2(P)$ decomposes as $h(e,y) = E[h \mid \varepsilon_{1t} = e] + (h(e,y) - E[h \mid \varepsilon_{1t} = e])$, where the first term lies in $L_2^0(f)$ and the second has conditional mean zero, so $\mathcal{T} = L_2^0(P)$. □

*Remark* A.4 (Relationship to the structural model). The bivariate marginal model is weakly larger than the set of bivariate marginals generated by the structural model (1)–(2), because stationarity and the structural recursion may restrict which pairs $(f,c)$ are jointly attainable. A larger model has a weakly larger tangent space and therefore a weakly higher efficiency bound. The bound $V_h^* = \mathrm{Var}(\psi_t^*)$ derived here should therefore be read as the efficiency bound for the enlarged bivariate marginal model. It is also an upper bound on the efficiency bound that would obtain from a smaller structural model that exploits additional restrictions. The paper does not rely on equality between the marginal and structural tangent spaces; establishing such equality would require a separate argument and is left for future work.

**Step 2. Pathwise derivative.** Under the submodel $P_\eta$, the marginal density and conditional mean become $f_\eta$ and $g_{h,\eta}$. The functional is

$$\theta(\eta) = \int g_{h,\eta}(e+\delta)\, f_\eta(e)\, de - \int g_{h,\eta}(e)\, f_\eta(e)\, de.$$

Differentiating at $\eta = 0$, using $f_\eta(e) = f(e)(1+\eta\, s_1(e)) + o(\eta)$ and the product rule:

$$\theta'(0) = \underbrace{\int g'_{h,\eta}(e+\delta)\, f(e)\, de - \int g'_{h,\eta}(e)\, f(e)\, de}_{(I)} + \underbrace{E\big[s_1(\varepsilon_{1t})\big(g_h(\varepsilon_{1t}+\delta) - g_h(\varepsilon_{1t})\big)\big]}_{(II)}, \tag{24}$$

where $g'_{h,\eta}(e) = \frac{d}{d\eta} \int y\, c_\eta(y \mid e)\, dy\big|_{\eta=0}$. Since $c_\eta(y \mid e) = c(y \mid e)(1+\eta\, s_2(e,y)) + o(\eta)$ and $E[s_2 \mid \varepsilon_{1t} = e] = 0$, this gives:

$$g'_{h,\eta}(e) = \int y\, s_2(e,y)\, c(y \mid e)\, dy = E\big[y_{t+h}\, s_2(e, y_{t+h}) \mid \varepsilon_{1t} = e\big]. \tag{25}$$

*Term (I).* Substituting $u = e + \delta$ in the first integral:

$$\begin{aligned}(I) &= \int g'_{h,\eta}(u)\, f(u-\delta)\, du - \int g'_{h,\eta}(e)\, f(e)\, de \\ &= \int g'_{h,\eta}(e)\, \big[f(e-\delta) - f(e)\big]\, de \\ &= E\big[g'_{h,\eta}(\varepsilon_{1t})\, (r_\delta(\varepsilon_{1t}) - 1)\big].\end{aligned} \tag{26}$$

Now apply (25) and the tower property:

$$
\begin{aligned}
(I) &= E\Big[E\big[y_{t+h}\, s_2(\varepsilon_{1t}, y_{t+h}) \mid \varepsilon_{1t}\big] \cdot (r_\delta(\varepsilon_{1t}) - 1)\Big] \\
&= E\big[y_{t+h}\, s_2(\varepsilon_{1t}, y_{t+h})\, (r_\delta(\varepsilon_{1t}) - 1)\big]. \qquad (27)
\end{aligned}
$$

Combining (27) and (II):

$$
\theta'(0) = E\big[s_1(\varepsilon_{1t})\big(g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t})\big)\big] + E\big[s_2(\varepsilon_{1t}, y_{t+h})\, (r_\delta(\varepsilon_{1t}) - 1)\, y_{t+h}\big]. \qquad (28)
$$

**Step 3. Verification of the Riesz representation.** I verify that $\psi_t^* = [g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t})] + (r_\delta(\varepsilon_{1t}) - 1)(y_{t+h} - g_h(\varepsilon_{1t})) - \theta$ satisfies $\theta'(0) = E[\psi_t^* \cdot s(\varepsilon_{1t}, y_{t+h})]$ for every score $s = s_1 + s_2 \in \mathcal{T}$. Since $\mathcal{T} = L_2^0(P)$, this identifies $\psi_t^*$ as the unique element of $\mathcal{T}$ representing the pathwise derivative, hence the efficient influence function.

Expand $E[\psi_t^* \cdot s]$ into three terms using the decomposition of $\psi_t^*$:

*(a) Regression term.*

$$
\begin{aligned}
&E\big[\big(g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t})\big)\big(s_1(\varepsilon_{1t}) + s_2(\varepsilon_{1t}, y_{t+h})\big)\big] \\
&\quad = E\big[\big(g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t})\big) s_1(\varepsilon_{1t})\big] + E\big[\big(g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t})\big)\, \underbrace{E[s_2 \mid \varepsilon_{1t}]}_{=0}\big] \\
&\quad = E\big[\big(g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t})\big) s_1(\varepsilon_{1t})\big]. \qquad (29)
\end{aligned}
$$

This matches term (II) in (28).

*(b) Augmentation term.*

$$
\begin{aligned}
&E\big[(r_\delta(\varepsilon_{1t}) - 1)(y_{t+h} - g_h(\varepsilon_{1t}))\big(s_1(\varepsilon_{1t}) + s_2(\varepsilon_{1t}, y_{t+h})\big)\big] \\
&\quad = E\big[(r_\delta - 1)\, \underbrace{E[y_{t+h} - g_h(\varepsilon_{1t}) \mid \varepsilon_{1t}]}_{=0} \cdot s_1(\varepsilon_{1t})\big] + E\big[(r_\delta(\varepsilon_{1t}) - 1)(y_{t+h} - g_h(\varepsilon_{1t}))\, s_2(\varepsilon_{1t}, y_{t+h})\big] \\
&\quad = E\big[(r_\delta(\varepsilon_{1t}) - 1)\, y_{t+h}\, s_2(\varepsilon_{1t}, y_{t+h})\big] - E\big[(r_\delta(\varepsilon_{1t}) - 1)\, g_h(\varepsilon_{1t})\, \underbrace{E[s_2 \mid \varepsilon_{1t}]}_{=0}\big] \\
&\quad = E\big[(r_\delta(\varepsilon_{1t}) - 1)\, y_{t+h}\, s_2(\varepsilon_{1t}, y_{t+h})\big]. \qquad (30)
\end{aligned}
$$

This matches term (I) in (28).

*(c) Centering term.* $E[\theta \cdot s] = \theta \cdot E[s] = 0$ because $s \in L_2^0(P)$.

Adding (29)–(30) recovers (28), confirming that

$$
\theta'(0) = E[\psi_t^* \cdot s(\varepsilon_{1t}, y_{t+h})] \quad \text{for all } s \in \mathcal{T}. \qquad (31)
$$

Since the tangent space equals $L_2^0(P)$, the function $\psi_t^*$ is the efficient influence function. The semiparametric efficiency bound is $V_h^* = \mathrm{Var}(\psi_t^*)$. □

*Remark* A.5 (The orthogonal decomposition). The verification reveals why the two components of $\psi_t^*$ are orthogonal (as recorded in Appendix A.4). The regression term $g_h(\varepsilon_{1t} + \delta) - g_h(\varepsilon_{1t})$ is measurable with respect to $\varepsilon_{1t}$ alone, while the augmentation term

$(r_\delta - 1)(y_{t+h} - g_h(\varepsilon_{1t}))$ has conditional mean zero given $\varepsilon_{1t}$. Their covariance therefore vanishes by the tower property, yielding the variance decomposition *(VD)*.

## A.4 Variance Decomposition and Plug-in Comparison

Let

$$m_h(e) \equiv g_h(e+\delta) - g_h(e), \qquad u_{t+h} \equiv y_{t+h} - g_h(\varepsilon_{1t}), \qquad \sigma^2(e) \equiv \mathrm{Var}(y_{t+h} \mid \varepsilon_{1t} = e).$$

Because $E[u_{t+h} \mid \varepsilon_{1t}] = 0$, the contemporaneous covariance between $m_h(\varepsilon_{1t})$ and $(r_\delta(\varepsilon_{1t}) - 1)u_{t+h}$ is zero. The marginal-model variance therefore decomposes as

$$V_h^* = \underbrace{\mathrm{Var}\big[m_h(\varepsilon_{1t})\big]}_{V_h^{\text{oracle}}} + \underbrace{E\Big[(r_\delta(\varepsilon_{1t}) - 1)^2 \sigma^2(\varepsilon_{1t})\Big]}_{V_h^{\text{aug}}}. \tag{VD}$$

The first term is the variance of the infeasible estimator that averages $m_h(\varepsilon_{1t})$ with $g_h$ treated as known. It is not generally the asymptotic variance of a feasible nonparametric regression estimator. For example, if $y_{t+h} = a + b\varepsilon_{1t} + u_{t+h}$, then $m_h(\varepsilon_{1t}) = b\delta$ and $V_h^{\text{oracle}} = 0$, but a feasible regression estimator must estimate $b$ and has positive sampling variance unless $u_{t+h} = 0$.

The second term is the residual-noise component generated by the Riesz representer of the functional $a \mapsto E[a(\varepsilon_{1t} + \delta) - a(\varepsilon_{1t})]$. That representer is $r_\delta(\varepsilon_{1t}) - 1$ because

$$E[a(\varepsilon_{1t} + \delta) - a(\varepsilon_{1t})] = E[(r_\delta(\varepsilon_{1t}) - 1)a(\varepsilon_{1t})].$$

A pure reweighting estimator based solely on (R) has score $y_{t+h}(r_\delta - 1) - \mathrm{ARF}_h(\delta)$ and generally adds unnecessary variation by weighting the conditional mean itself; this is why reweighting is used here as an augmentation device rather than as the primary estimator.

The comparison with a feasible regression plug-in should be read narrowly. If a regression plug-in is tuned so that

$$\widehat{\theta}_h^{\text{reg}} - \mathrm{ARF}_h(\delta) = \frac{1}{n_h} \sum_{t=1}^{n_h} \psi_t^* + o_p(n_h^{-1/2}),$$

then it is first-order equivalent to the block cross-fitted DR estimator under Proposition A.2. This observation is only a benchmark: conventional regression-only nonparametric LPs may have smoothing or misspecification bias, whereas the DR estimator removes first-order regression bias whenever the density-ratio arm is correct.

## A.5 Consistency of the Block Cross-Fitted DR Estimator

**Proposition A.1** (Double robustness of the block cross-fitted estimator)**.** *Under Assumptions A.1–A.6 and Assumption A.5(a),*

$$\widehat{\mathrm{ARF}}_h^{BCF}(\delta) \xrightarrow{p} \mathrm{ARF}_h(\delta)$$

*as $T \to \infty$. The boundedness requirement on the misspecified component is part of Assumption A.5(a); trimming and capping provide finite-sample envelopes but do not remove the need to control diverging caps asymptotically.*

*Proof.* Let

$$\phi(W_{t,h}; g, r) = g(\varepsilon_{1t} + \delta) - g(\varepsilon_{1t}) + \{r(\varepsilon_{1t}) - 1\}\{y_{t+h} - g(\varepsilon_{1t})\},$$

where $W_{t,h} = (\varepsilon_{1t}, y_{t+h})$. For $t \in B_{k,h}$ write $\hat{\eta}_k = (\hat{g}_{k,h}, \hat{r}_{k,\delta})$ and $\eta_0 = (g_h, r_\delta)$. The block cross-fitted estimator satisfies

$$\widehat{\mathrm{ARF}}_h^{BCF} - \mathrm{ARF}_h = \frac{1}{n_h}\sum_{t=1}^{n_h} \psi_t^* + \frac{1}{n_h}\sum_{k=1}^{K}\sum_{t\in B_{k,h}} \{\phi(W_{t,h}; \hat{\eta}_k) - \phi(W_{t,h}; \eta_0)\}.$$

Conditioning on the training sigma-field for fold $k$, the population drift of the second term is

$$P\{\phi(W; \hat{\eta}_k) - \phi(W; \eta_0)\} = -P\{(\hat{r}_{k,\delta} - r_\delta)(\hat{g}_{k,h} - g_h)\},$$

where the density-ratio identity converts $P[\hat{g}_{k,h}(E + \delta) - \hat{g}_{k,h}(E)]$ into $P[(r_\delta(E) - 1)\hat{g}_{k,h}(E)]$. Thus the drift is bounded by

$$\|\hat{r}_{k,\delta} - r_\delta\|_{2,f}\, \|\hat{g}_{k,h} - g_h\|_{2,f}.$$

Under Assumption A.5(a), this bound is $o_p(1)$ if either nuisance arm is consistently estimated and the other remains $O_p(1)$. Lemma A.1 gives the corresponding centered block empirical-process term as $o_p(1)$ under the same consistency conditions. Finally, $n_h^{-1}\sum_t \psi_t^* = o_p(1)$ by the weak-dependence law of large numbers. □

## A.6 Asymptotic Normality

**Proposition A.2** (Asymptotic normality of the block cross-fitted DR estimator)**.** *Under Assumptions A.1–A.6 and Assumption A.5(b),*

$$\sqrt{n_h}\big(\widehat{\mathrm{ARF}}_h^{BCF}(\delta) - \mathrm{ARF}_h(\delta)\big) \xrightarrow{d} N(0, \Sigma_h^*) \tag{32}$$

*where*

$$\Sigma_h^* = \sum_{j=-\infty}^{\infty} \mathrm{Cov}(\psi_t^*, \psi_{t-j}^*).$$

*The contemporaneous variance $V_h^* = \mathrm{Var}(\psi_t^*)$ is recovered only when the nonzero-lag autocovariances vanish. Finite lag order in the structural recursion does not by itself imply this condition. If Assumption A.7 also holds, the long-run variance estimator based on the estimated block cross-fitted scores is consistent for $\Sigma_h^*$.*

Under Assumption A.7, the long-run variance can be estimated by a Newey–West HAC estimator

$$\hat{\Sigma}_h = \hat{\gamma}_0 + \sum_{j=1}^{B_T} \kappa\left(\frac{j}{B_T}\right)(\hat{\gamma}_j + \hat{\gamma}_j'), \qquad \hat{\gamma}_j = \frac{1}{n_h}\sum_{t=j+1}^{n_h} \tilde{\phi}_{t,h}^{BCF}\,\tilde{\phi}_{t-j,h}^{BCF}, \tag{33}$$

where $\tilde{\phi}_{t,h}^{BCF} = \hat{\phi}_{t,h}^{BCF} - \widehat{\mathrm{ARF}}_h^{BCF}(\delta)$, $\kappa$ is the Bartlett kernel, and $B_T \to \infty$ with $B_T/T \to 0$. A fixed bandwidth proportional only to the horizon is justified only under a genuine finite-dependence assumption, which is stronger than finite lag order in the structural equations.

**Lemma A.3** (HAC replacement for estimated scores)**.** *Let*

$$\hat{q}_{t,h} = \hat{\phi}_{t,h}^{BCF} - \widehat{\mathrm{ARF}}_h^{BCF}(\delta), \qquad q_{t,h}^* = \phi_{t,h}^* - \mathrm{ARF}_h(\delta) = \psi_t^*.$$

*Let $\hat{\Sigma}_h(\hat{q})$ denote the HAC estimator computed from $\{\hat{q}_{t,h}\}$ and $\hat{\Sigma}_h(q^*)$ the corresponding infeasible HAC estimator computed from $\{q_{t,h}^*\}$. Under Assumption A.7,*

$$\hat{\Sigma}_h(\hat{q}) - \hat{\Sigma}_h(q^*) = o_p(1).$$

*Consequently, if the infeasible HAC estimator based on $q_{t,h}^*$ is consistent under Assumption A.4, then the feasible HAC estimator based on $\hat{\phi}_{t,h}^{BCF}$ is also consistent.*

*Proof.* Write $d_{t,h} = \hat{\phi}_{t,h}^{BCF} - \phi_{t,h}^*$ and $\bar{d}_h = n_h^{-1}\sum_{t=1}^{n_h} d_{t,h}$, so that

$$\hat{q}_{t,h} = q_{t,h}^* + (d_{t,h} - \bar{d}_h).$$

Let $\|a\|_n^2 = n_h^{-1}\sum_{t=1}^{n_h} a_t^2$. Since $\|d - \bar{d}\|_n \le 2\|d\|_n$, Assumption A.7 gives $B_T\|d - \bar{d}\|_n = o_p(1)$. For each lag $j \le B_T$, Cauchy–Schwarz yields

$$|\hat{\gamma}_j(\hat{q}) - \hat{\gamma}_j(q^*)| \le 2\|q^*\|_n\|d - \bar{d}\|_n + \|d - \bar{d}\|_n^2 + o_p(B_T^{-1}),$$

where the endpoint difference from using $n_h - j$ rather than $n_h$ observations is absorbed because $B_T/n_h \to 0$ and $E[(q_{t,h}^*)^2] < \infty$. The weak-dependence and moment conditions imply $\|q^*\|_n = O_p(1)$. Summing the preceding bound over $j \le B_T$ with bounded kernel weights gives

$$|\hat{\Sigma}_h(\hat{q}) - \hat{\Sigma}_h(q^*)| \le CB_T\{O_p(1)\|d - \bar{d}\|_n + \|d - \bar{d}\|_n^2\} + o_p(1) = o_p(1).$$

This proves the replacement claim. □

*Proof of Proposition A.2.* By Lemma A.1,

$$\widehat{\mathrm{ARF}}_h^{BCF} - \mathrm{ARF}_h = \frac{1}{n_h}\sum_{t=1}^{n_h}\psi_t^* - \sum_{k=1}^{K}\frac{|B_{k,h}|}{n_h}P\{(\hat{r}_{k,\delta} - r_\delta)(\hat{g}_{k,h} - g_h)\} + o_p(n_h^{-1/2}).$$

The product-bias term is bounded in absolute value by

$$\max_k \|\hat{r}_{k,\delta} - r_\delta\|_{2,f}\,\|\hat{g}_{k,h} - g_h\|_{2,f} = o_p(n_h^{-1/2})$$

under Assumption A.5(b). Therefore

$$\sqrt{n_h}\big(\widehat{\mathrm{ARF}}_h^{BCF} - \mathrm{ARF}_h\big) = \frac{1}{\sqrt{n_h}}\sum_{t=1}^{n_h}\psi_t^* + o_p(1).$$

The stated weak-dependence condition and moment assumptions yield a central limit theorem for the partial sum of $\psi_t^*$ with long-run variance $\Sigma_h^*$ (e.g., Davidson, 1994, Theorem 27.4). Standard kernel long-run-variance results give consistency of the infeasible HAC estimator computed from $\{\psi_t^*\}$ under the same dependence conditions (Newey and West, 1987; Andrews, 1991). Lemma A.3 then transfers this consistency to the feasible HAC estimator computed from the estimated block cross-fitted scores when Assumption A.7 holds. □

*Remark* A.6 (Boundedness of the misspecified component). The Cauchy–Schwarz step in the proof requires that the $L_2$ norm of the misspecified nuisance component is $O_p(1)$; consistency of the well-specified component alone does not control the product if the other component diverges. Capping the density ratio at $M_N$ and winsorizing the regression at $L_N$ provide finite-sample envelopes. If the caps are fixed, boundedness is immediate but the estimator targets a stabilized version of the nuisance problem. If the caps diverge with the fold training size so that the original ARF remains the target, boundedness must either be verified from the data-generating process or imposed as the high-level condition in Assumption A.5(a). This is why the formal double-robustness statement separates consistency of one arm from envelope control of the other.

# B Primitive Rate Conditions

This appendix supplies one sufficient primitive route for the product-rate condition in Assumption A.5(b). It is not needed for identification, double robustness, or the influence-function expansion, all of which rely on the high-level conditions in Appendix A. The conditions below verify the relevant fold-specific population norms for trimmed and capped kernel-based nuisance estimators; empirical-process control is supplied separately by Assumption A.6 and Lemma A.1.

**Assumption B.1** (Smoothness and tail/log-ratio control)**.** (a) The density $f$ satisfies $f \in C^2(\mathbb{R})$ with $f, f', f'' \in L_1(\mathbb{R}) \cap L_2(\mathbb{R})$ and $\sup_e |e^k f^{(j)}(e)| < \infty$ for $j, k \in \{0, 1, 2\}$. (b) The density has two-sided Gaussian-type tail control on the effective support: for sufficiently large $|e|$, $c_1 \exp(-C_1 e^2) \leq f(e) \leq c_2 \exp(-C_2 e^2)$ for positive constants $c_1, c_2, C_1, C_2$, and $|\partial_e \log f(e)| \leq C(1 + |e|)$ wherever $f(e) > 0$. (c) The conditional mean $g_h \in C^2(\mathbb{R})$ with $\int (g_h''(e))^2 f(e)\, de < \infty$.

**Assumption B.2** (Moments)**.** $E[y_{t+h}^4] < \infty$, $E[g_h(\varepsilon_{1t})^4] < \infty$, and $E[r_\delta(\varepsilon_{1t})^4] < \infty$.

**Assumption B.3** (Conditional variance integrability)**.** The conditional variance $\sigma^2(e) \equiv \mathrm{Var}(y_{t+h} \mid \varepsilon_{1t} = e)$ is bounded: $\sup_e \sigma^2(e) < \infty$.

**Assumption B.4** (Kernel and bandwidth)**.** The kernel $K$ is a bounded, symmetric, second-order kernel with $\int K = 1$, $\mu_2 \equiv \int u^2 K(u)\, du \neq 0$, $R(K) \equiv \int K^2(u)\, du < \infty$, and $K(u) > 0$ for all $u$ (e.g., the Gaussian kernel). For each fold, with training size $N_{k,h} = |I_{k,h}|$, the bandwidths satisfy $b_{g,k} \asymp N_{k,h}^{-1/5} (\log N_{k,h})^{1/10}$ and $b_{f,k} \asymp N_{k,h}^{-1/5} (\log N_{k,h})^{1/10}$.

The rate calculation controls population norms of the nuisance errors for the kernel implementation. It is phrased as an interior one-dimensional kernel rate plus explicit tail and clipping remainders. For unbounded shocks, the effective-support and clipping remainders must be checked, imposed, or interpreted as defining a stabilized target.

**Trimmed and capped estimators.** For fold $k$, let $N_{k,h} = |I_{k,h}|$. Let $c_N \to \infty$ be a deterministic effective-support radius and define

$$\mathcal{S}_N = \{e : |e| \leq c_N,\ |e - \delta| \leq c_N\}.$$

Let

$$a_N = \frac{1}{2} \inf_{e \in \mathcal{S}_N} f(e),$$

or more generally any positive sequence of the same order as the lower density bound on $\mathcal{S}_N$ and satisfying $\sup_{e \in \mathcal{S}_N} |\hat{f}_k(e) - f(e)| = o_p(a_N)$. This condition ensures that the denominator of the plug-in density ratio is stable on the effective support. Let $\hat{f}_k$ be a kernel density estimator of $f$ computed from $\{\varepsilon_{1s} : s \in I_{k,h}\}$ with bandwidth $b_{f,k}$, and define the clipping map

$$\Pi_{[-L_N, L_N]}(x) = \min\{\max(x, -L_N), L_N\}.$$

The formal regression nuisance used in the rate calculation is

$$\hat{g}^T_{k,h}(e) = \Pi_{[-L_N,L_N]}\{\hat{g}_{k,h}(e)\}\,\mathbb{I}\{e \in \mathcal{S}_N,\ \hat{f}_k(e) \geq a_N\},$$

where $L_N \to \infty$ is a regression envelope. The formal density-ratio nuisance is

$$\hat{r}^T_{k,\delta}(e) = \frac{\hat{f}_k(e-\delta)}{\hat{f}_k(e)}\mathbb{I}\left\{e \in \mathcal{S}_N,\ \hat{f}_k(e) \geq a_N,\ \left|\frac{\hat{f}_k(e-\delta)}{\hat{f}_k(e)}\right| \leq M_N\right\} + 1 \cdot \mathbb{I}\{\text{otherwise}\},$$

where $M_N \to \infty$ caps extreme ratio estimates. The default value 1 outside the stable region ensures that $\hat{r}^T_{k,\delta} - 1 = 0$ there, so trimmed observations do not contribute to the augmentation term.

Define the regression tail and winsorization remainder

$$\tau_{g,N} = \|g_h\mathbb{I}\{\mathcal{S}_N^c\}\|_{2,f} + \|g_h\mathbb{I}\{\mathcal{S}_N^c\}\|_{2,f_\delta} + \|(|g_h| - L_N)_+\|_{2,f} + \|(|g_h| - L_N)_+\|_{2,f_\delta},$$

and the density-ratio tail and clipping remainder

$$\tau_{r,N} = \|(r_\delta - 1)\mathbb{I}\{\mathcal{S}_N^c\}\|_{2,f} + \|(r_\delta - 1)\mathbb{I}\{\mathcal{S}_N^c\}\|_{2,\sigma} + \|(|r_\delta| - M_N)_+\|_{2,f} + \|(|r_\delta| - M_N)_+\|_{2,\sigma}.$$

Here $(x)_+ = \max\{x, 0\}$ and the $\sigma$-weighted terms are omitted when the bounded-variance condition in Assumption B.3 is used to reduce them to unweighted $L_2(P_f)$ terms.

**Proposition B.1** (Fold-specific population-rate verification with explicit tails)**.** *Suppose Assumptions A.1–B.4 hold, with the strengthened tail/log-ratio condition in Assumption B.1. Let $\delta$ be fixed as $T \to \infty$, $K$ fixed, and $N_{k,h} = |I_{k,h}| \asymp n_h$ uniformly over folds. Choose $c_N$, $a_N$, $M_N$, and $L_N$ as above. Then, uniformly over $k = 1, \ldots, K$,*

$$\|\hat{g}^T_{k,h} - g_h\|_{2,f} + \|\hat{g}^T_{k,h} - g_h\|_{2,f_\delta} = O_p\{n_h^{-2/5}(\log n_h)^{1/4}\} + O(\tau_{g,N}), \tag{34}$$

$$\|\hat{r}^T_{k,\delta} - r_\delta\|_{2,f} + \|\hat{r}^T_{k,\delta} - r_\delta\|_{2,\sigma} = O_p\{n_h^{-2/5+o(1)}(\log n_h)^{1/4}\} + O(\tau_{r,N}). \tag{35}$$

*Consequently, the product-rate condition in Assumption A.5(b) holds whenever*

$$\left(n_h^{-2/5}(\log n_h)^{1/4} + \tau_{g,N}\right)\left(n_h^{-2/5+o(1)}(\log n_h)^{1/4} + \tau_{r,N}\right) = o(n_h^{-1/2}).$$

*In the special case in which $\tau_{g,N} = o\{n_h^{-2/5}(\log n_h)^{1/4}\}$ and $\tau_{r,N} = o\{n_h^{-2/5+o(1)}(\log n_h)^{1/4}\}$, this reduces to the simpler product rate*

$$\max_k \|\hat{g}^T_{k,h} - g_h\|_{2,f}\,\|\hat{r}^T_{k,\delta} - r_\delta\|_{2,f} = O_p\{n_h^{-4/5+o(1)}(\log n_h)^{1/2}\} = o_p(n_h^{-1/2}).$$

*When the shock density $f$ is known up to a finite-dimensional parameter $\theta$ estimated at the $\sqrt{n_h}$ rate within each training fold, the interior part of (35) improves to $O_p(n_h^{-1/2})$ under standard smoothness of $r_\delta(\cdot;\theta)$.*

*Proof.* The proof is a rate calculation for population norms within a generic fold. Because

$K$ is fixed and $N_{k,h} \asymp n_h$ uniformly over folds, the same orders hold uniformly over $k$. Empirical-process control is supplied separately by Lemma A.1.

*Regression rate.* On $\mathcal{S}_N$, the density is bounded below by construction and the local-linear denominator is stable on the event $\sup_{e\in\mathcal{S}_N} |\hat{f}_k(e) - f(e)| = o_p(a_N)$. Standard local-linear regression theory under the weak-dependence and moment conditions gives the integrated interior bound

$$O_p(b_{g,k}^2) + O_p\Big((N_{k,h} b_{g,k})^{-1/2} c_N^{1/2}\Big)$$

in $L_2$ norm over the effective support, with lower-order covariance terms controlled by Assumption A.4. For $c_N = O(\sqrt{\log n_h})$ and $b_{g,k} \asymp N_{k,h}^{-1/5}(\log N_{k,h})^{1/10}$, this is bounded by $O_p\{n_h^{-2/5}(\log n_h)^{1/4}\}$. The observations outside $\mathcal{S}_N$ and the winsorized part contribute at most $\tau_{g,N}$ by definition, yielding (34).

*Density-ratio rate.* On the stable interior, linearize the plug-in ratio:

$$\hat{r}_\delta(e) - r_\delta(e) = \frac{\Delta\hat{f}_k(e-\delta) - r_\delta(e)\Delta\hat{f}_k(e)}{f(e)} + \text{higher-order terms}, \qquad \Delta\hat{f}_k = \hat{f}_k - f.$$

The kernel density estimator satisfies the usual one-dimensional mean-square bound

$$E\{\Delta\hat{f}_k(u)^2\} = O\{b_{f,k}^4 (f''(u))^2\} + O\{f(u)/(N_{k,h} b_{f,k})\}.$$

The strengthened tail/log-ratio condition gives, uniformly over $\mathcal{S}_N$ with $c_N = O(\sqrt{\log n_h})$,

$$\sup_{e\in\mathcal{S}_N} r_\delta(e)^2 \le \exp\{O(|\delta|(1 + c_N + |\delta|))\} = n_h^{o(1)}$$

for fixed $\delta$. Integrating the linearized error over the effective support gives the interior squared-norm order $O_p\{n_h^{-4/5+o(1)}(\log n_h)^{1/2}\}$. The higher-order term is of smaller order on the denominator-stability event. The observations outside $\mathcal{S}_N$ and the ratio-clipped part contribute at most $\tau_{r,N}$ by definition. Taking square roots yields (35).

Multiplying the total regression and ratio bounds gives the stated sufficient condition for Assumption A.5(b). The parametric density-ratio statement follows from the delta method applied to $r_\delta(e;\theta)$ inside each training fold. □

*Remark* B.1 (Role of the tail/log-ratio condition). Assumption B.1(b) is stronger than an upper tail bound. The lower tail bound controls the denominator density on the effective support, and the log-ratio bound controls $f(e-\delta)/f(e)$ uniformly over the region where the kernel density estimator is used. This is the condition needed for the plug-in density-ratio proof. A one-sided sub-Gaussian upper bound alone is insufficient, because the denominator $f(e)$ may still be too small or irregular to control the ratio.

The primitive rate verification is only one sufficient route. The double-robustness property (Proposition A.1) and the efficient influence-function calculation (Proposition 3.2) require the high-level moment and overlap conditions, not this particular kernel-tail argument.

*Remark* B.2 (Fixed shifts and large-shift overlap). Proposition B.1 treats $\delta$ as fixed. Under Assumption B.1(b), the ratio bound on the deterministic effective support is $n_h^{o(1)}$ for any fixed shift, so no additional local-to-zero restriction on $|\delta|/\sigma_1$ is needed for the interior kernel calculation. The tail and clipping contributions remain explicit through $\tau_{g,N}$ and $\tau_{r,N}$; they may bind in finite samples or under aggressive effective-support choices. Large $|\delta|$ can still produce unstable weights and poor overlap. Regularized or clipped density-ratio estimators can stabilize finite-sample behavior; for the original untrimmed ARF, any fixed cap must diverge with $T$, otherwise the estimand is a trimmed or stabilized version of the ARF. Within the DR framework, density-ratio bias enters through the product with the regression error.

# C Implementation Details

## C.1 Buffer Length, Fold Geometry, and HAC Bandwidth

A practical default buffer is

$$m_T = \max\{h, \lceil n_h^{1/4} \rceil\},$$

or, when estimating a full path $h = 0, \ldots, H$ with a common fold geometry,

$$m_T = \max\{H, \lceil (T-H)^{1/4} \rceil\}.$$

The exponent is not essential. The formal conditions require $m_T \to \infty$, $m_T/n_h \to 0$, and sufficiently fast decay of the mixing coefficients at the buffer distance. Small fixed choices such as $m_T = h$ are useful sensitivity checks but are not the asymptotic default.

The same fold geometry can be used for all horizons by working on the common base sample $\{1, \ldots, T-H\}$. This sacrifices the final $H-h$ observations for shorter horizons but simplifies joint inference for the full impulse-response path. For joint bands, form the vector of cross-fitted scores across horizons and apply a vector HAC estimator.

Asymptotic normality (Proposition A.2) and the estimated-score replacement condition in Assumption A.7 imply that valid confidence intervals can be constructed from a HAC estimator of the long-run variance of $\hat{\phi}_{t,h}^{BCF}$. A finite lag order in the structural recursion does not by itself imply finite dependence of the score; in autoregressive systems the score can have serial autocovariances at arbitrarily long lags. Accordingly, the HAC bandwidth should follow a standard sequence with $B_T \to \infty$ and $B_T/T \to 0$ under weak dependence, with the additional restriction that the estimated-score error is small relative to $B_T^{-1}$.

## C.2 Density-Ratio Estimation

The main text only requires a fold-specific estimate of $r_\delta(e) = f(e-\delta)/f(e)$ trained on the same buffer-excluded observations as the outcome regression.

**Plug-in kernel density ratios.** Estimate $f$ from the training shocks $\{\varepsilon_{1s} : s \in I_{k,h}\}$ using a standard kernel density estimator $\hat{f}_k$, then form

$$\hat{r}_{k,\delta}(e) = \frac{\hat{f}_k(e-\delta)}{\hat{f}_k(e)}.$$

This approach inherits standard one-dimensional density-estimation theory. Its main practical weakness is tail instability when $\hat{f}_k(e)$ is small. The formal trimmed and capped version in Appendix B controls this instability through fold-specific density floors and ratio caps. For consistency for the original, untrimmed ARF when the true ratio is unbounded, the floor must vanish and the cap must diverge with the fold training size; a fixed cap instead targets a stabilized estimand.

**Parametric plug-in density ratios.** When the marginal shock density is credibly modeled by a low-dimensional family $f_\eta$, estimate $\eta$ on each fold's training shocks and set

$$\hat{r}_{k,\delta}(e) = \frac{f_{\hat{\eta}_k}(e-\delta)}{f_{\hat{\eta}_k}(e)}.$$

This is the approach used in the Monte Carlo exercises, where the shocks are Gaussian by construction. For a mean-zero Gaussian shock with variance $\sigma_1^2$,

$$r_\delta(e) = \exp\left(\frac{\delta e}{\sigma_1^2} - \frac{\delta^2}{2\sigma_1^2}\right),$$

with $\sigma_1^2$ replaced by its training-sample estimate in feasible block cross-fitted implementation. The parametric route is attractive when the first-stage structural shock normalization makes the marginal distribution plausible and stable, but it gives up the nonparametric robustness of the kernel plug-in ratio if the shock density is misspecified.

# D Monte Carlo Figures

Computations were performed using the Wake Forest University High Performance Computing Facility (Information Systems and Wake Forest University, 2021).

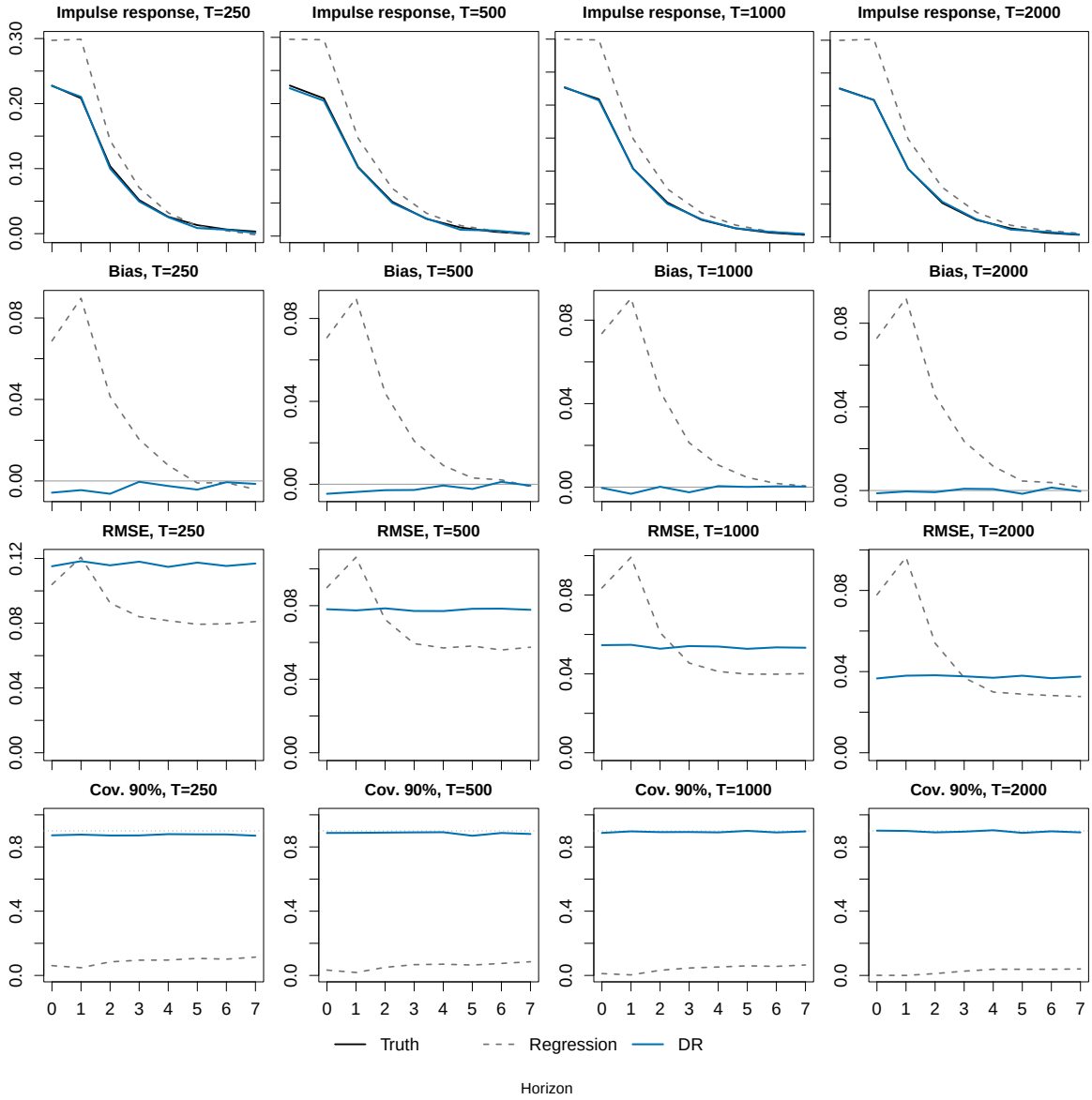


**Figure 1:** Misspecified regression. DGP with ReLU nonlinearity $q(x) = \max(x, 0)$ and shock size $\delta = 1$. Each column corresponds to a sample size $T$; rows show the estimated ARF, mean bias, RMSE, and 90% coverage. Dashed: linear LP omitting the nonlinear term; solid blue: DR estimator pairing the same misspecified regression with a parametric density ratio.

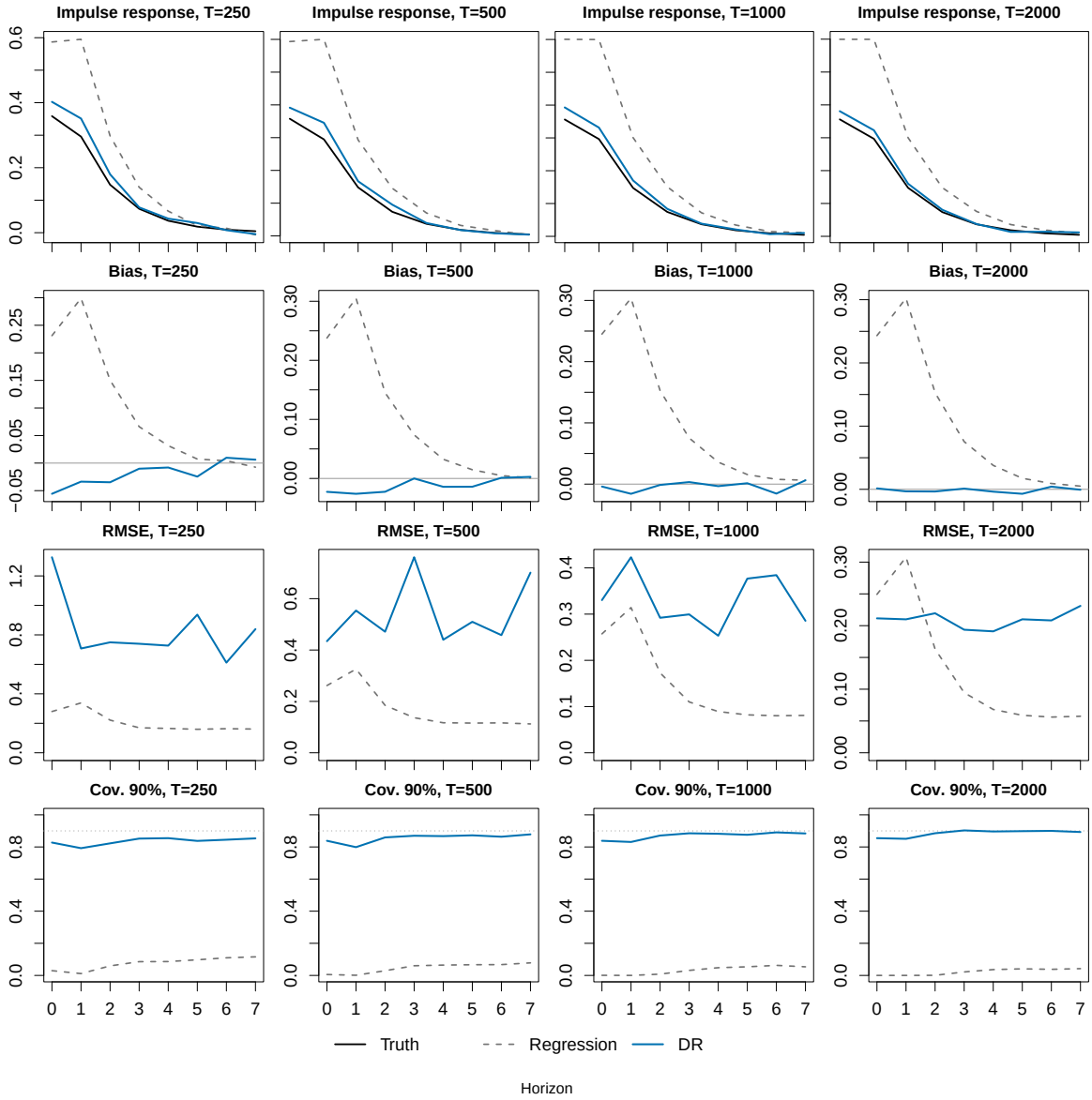


**Figure 2:** Misspecified regression. DGP with ReLU nonlinearity $q(x) = \max(x, 0)$ and shock size $\delta = 2$. Each column corresponds to a sample size $T$; rows show the estimated ARF, mean bias, RMSE, and 90% coverage. Dashed: linear LP omitting the nonlinear term; solid blue: DR estimator pairing the same misspecified regression with a parametric density ratio.

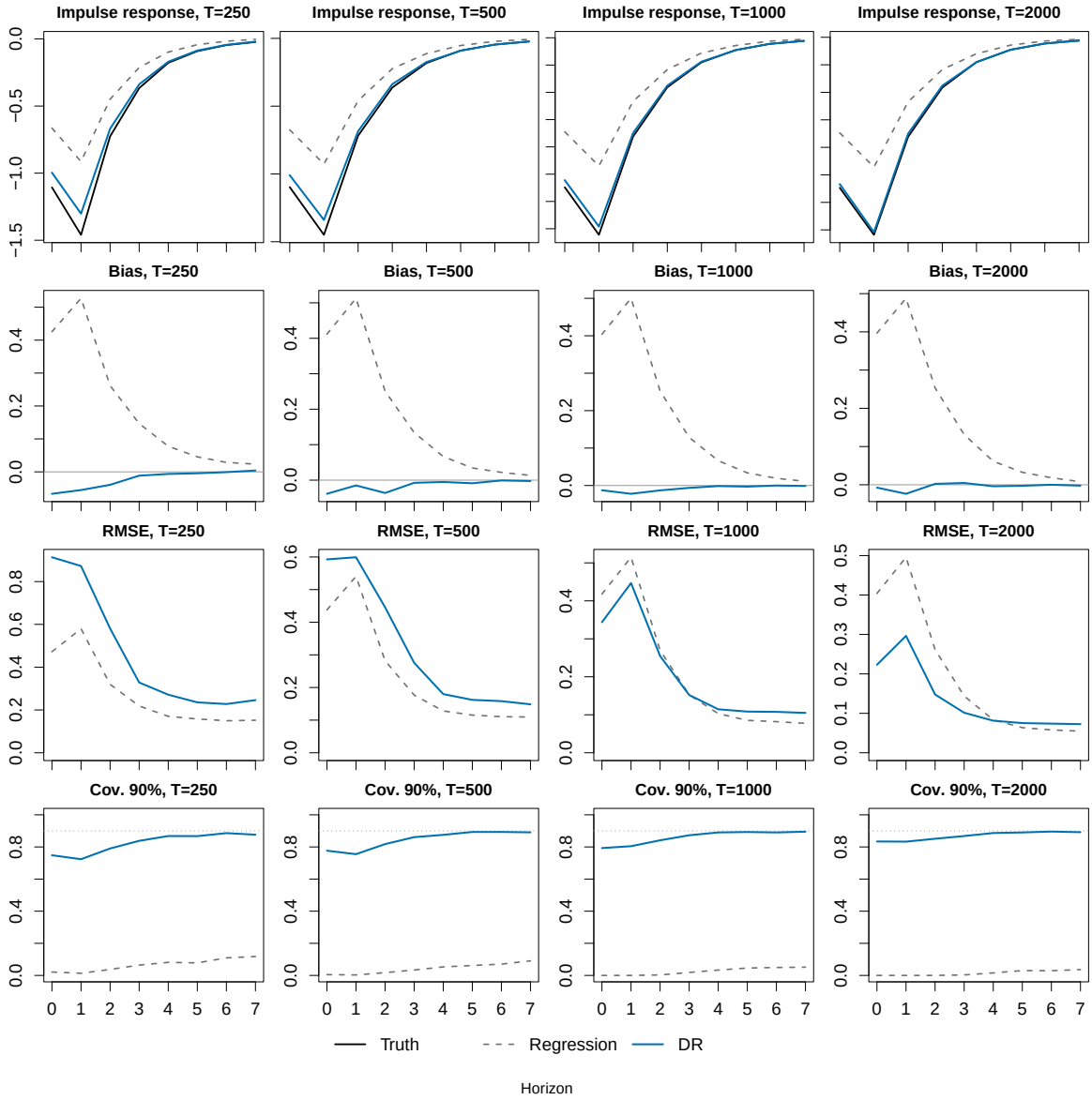


**Figure 3:** Misspecified regression. DGP with cubic nonlinearity $q(x) = x^3$ and shock size $\delta = 1$. Each column corresponds to a sample size $T$; rows show the estimated ARF, mean bias, RMSE, and 90% coverage. Dashed: linear LP omitting the nonlinear term; solid blue: DR estimator pairing the same misspecified regression with a parametric density ratio.

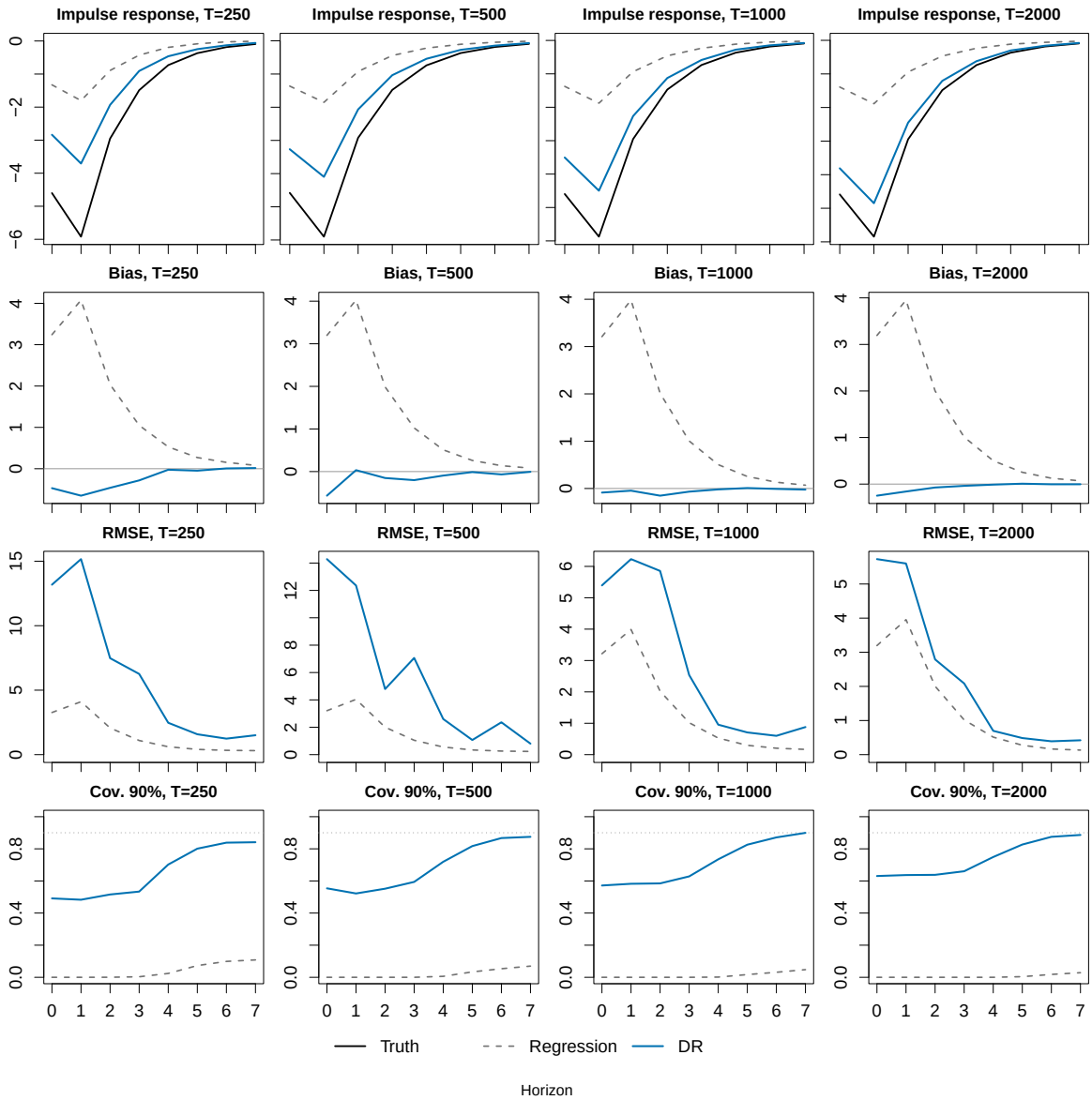


**Figure 4:** Misspecified regression. DGP with cubic nonlinearity $q(x) = x^3$ and shock size $\delta = 2$. Each column corresponds to a sample size $T$; rows show the estimated ARF, mean bias, RMSE, and 90% coverage. Dashed: linear LP omitting the nonlinear term; solid blue: DR estimator pairing the same misspecified regression with a parametric density ratio.

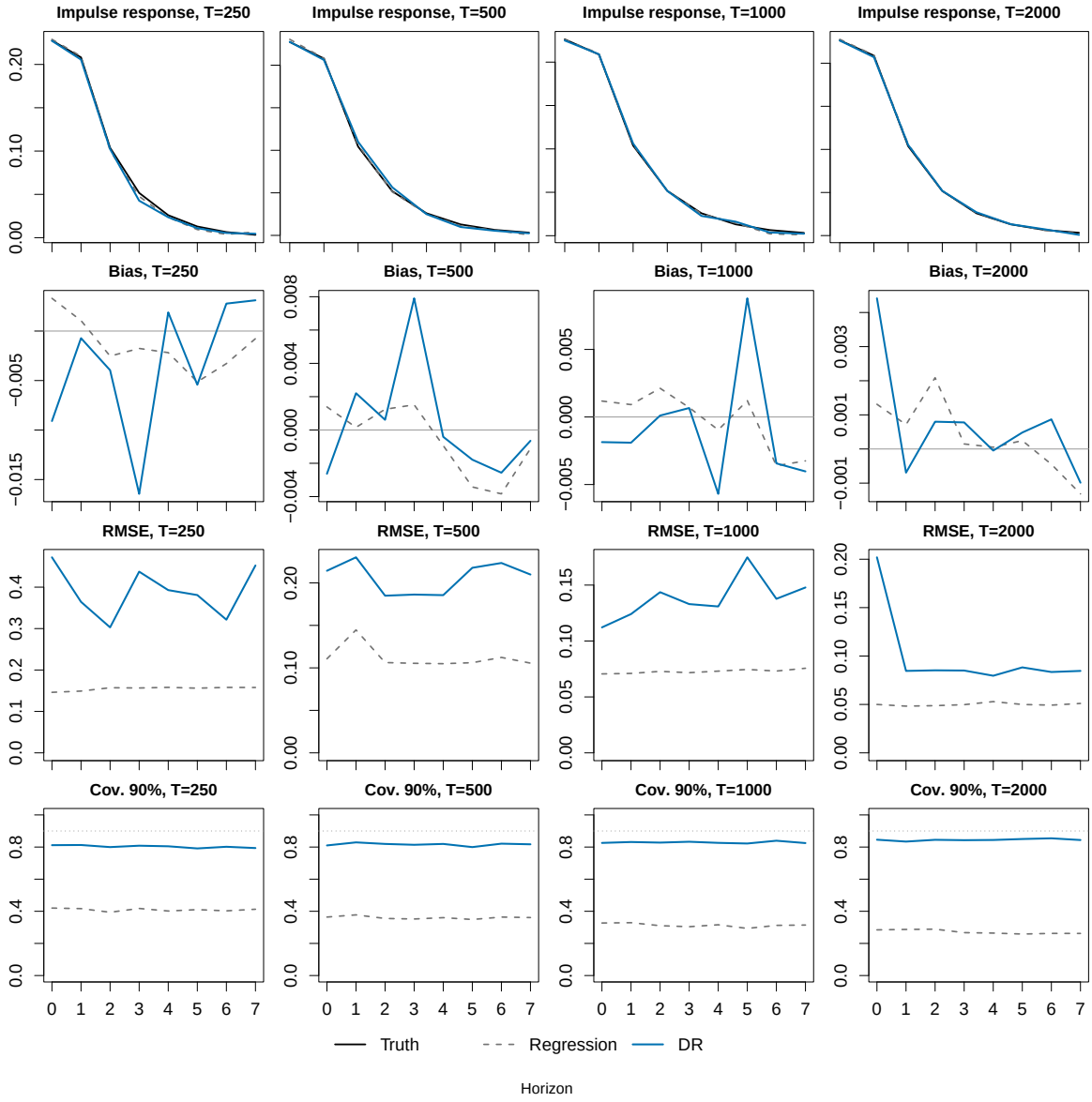


**Figure 5:** Flexibly estimated regression. DGP with ReLU nonlinearity $q(x) = \max(x, 0)$ and shock size $\delta = 1$. Each column corresponds to a sample size $T$; rows show the estimated ARF, mean bias, RMSE, and 90% coverage. Dashed: local linear LP; solid blue: DR estimator pairing the same flexible regression with a parametric density ratio.

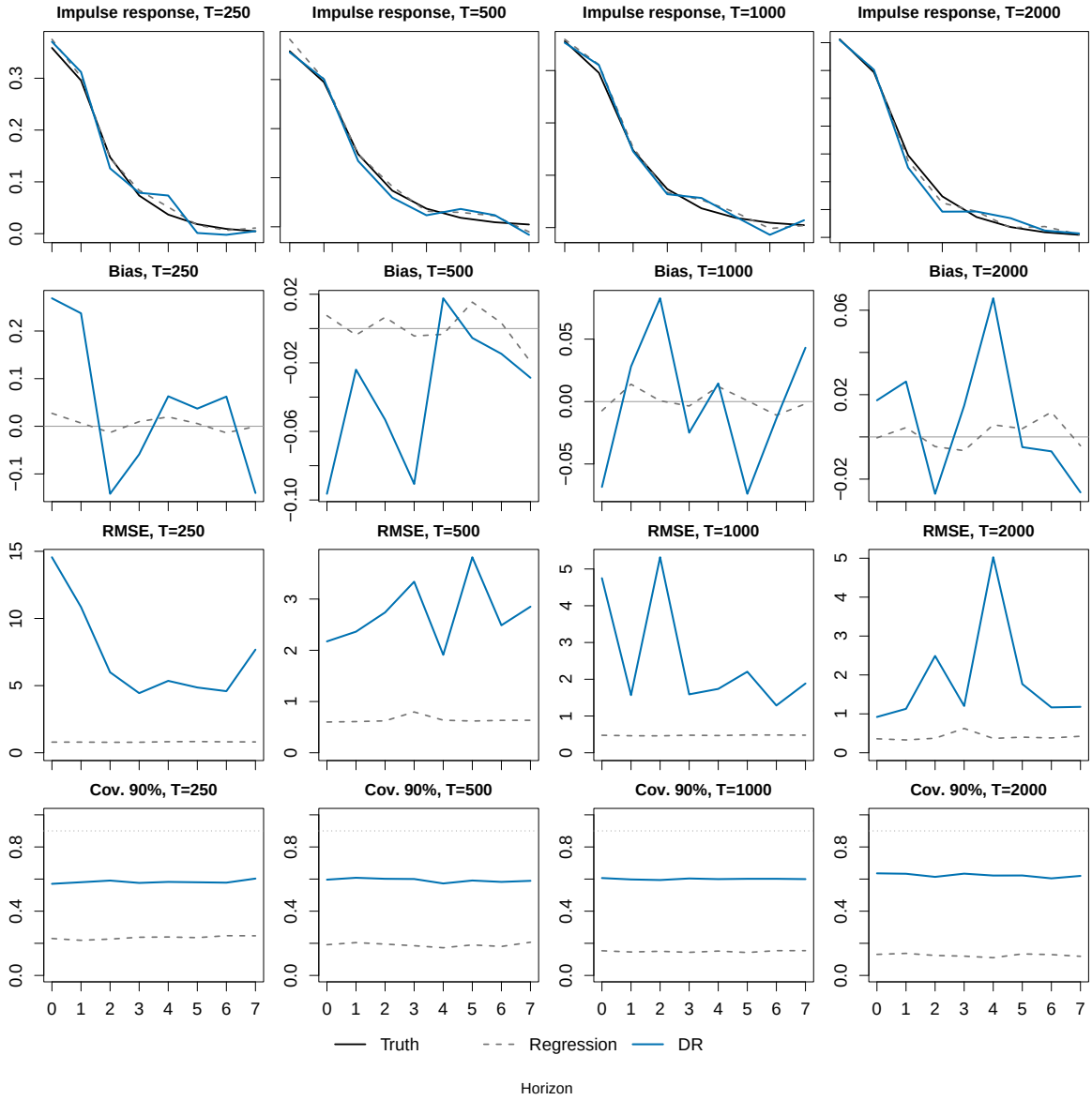


**Figure 6:** Flexibly estimated regression. DGP with ReLU nonlinearity $q(x) = \max(x, 0)$ and shock size $\delta = 2$. Each column corresponds to a sample size $T$; rows show the estimated ARF, mean bias, RMSE, and 90% coverage. Dashed: local linear LP; solid blue: DR estimator pairing the same flexible regression with a parametric density ratio.

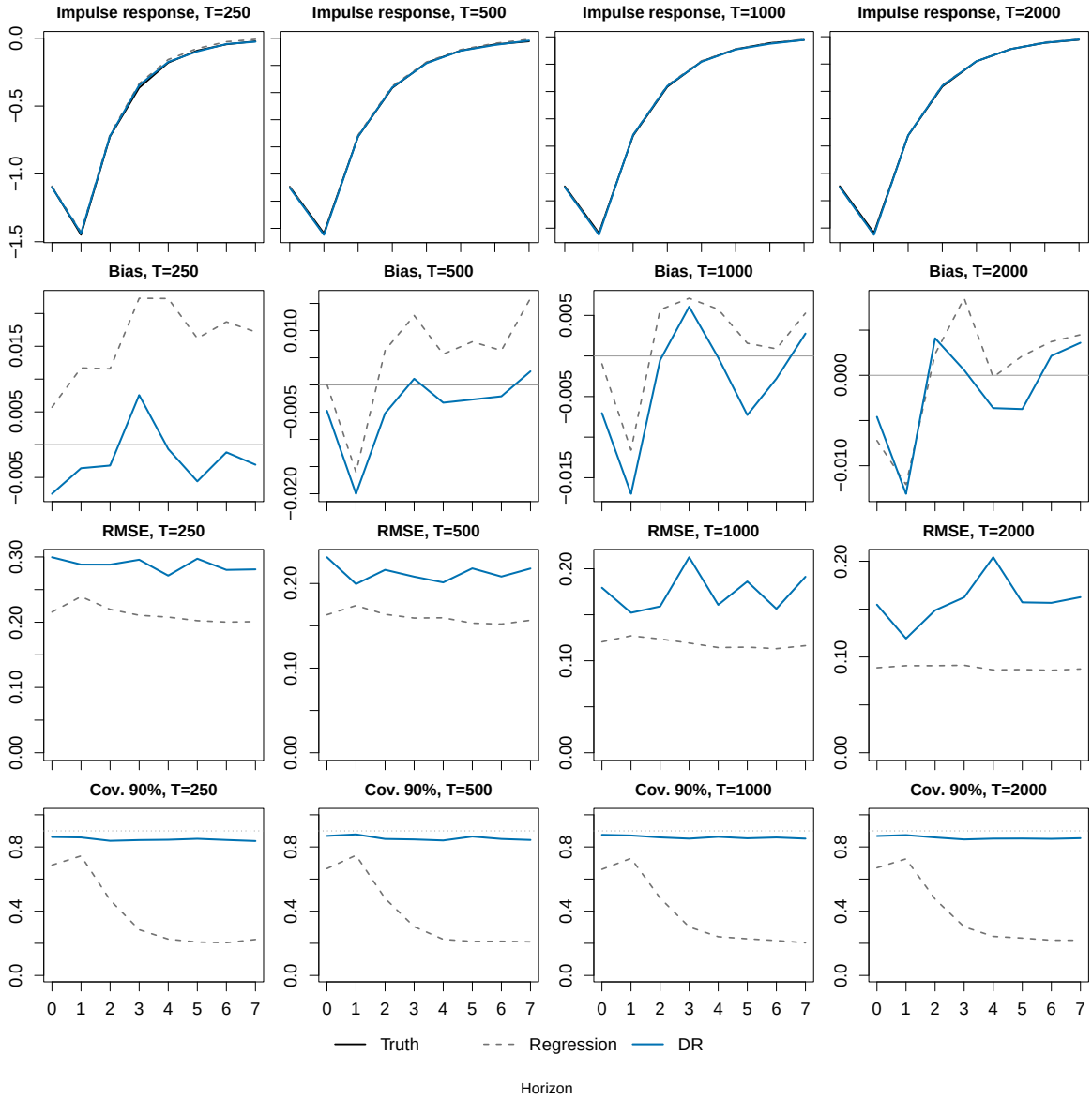


**Figure 7:** Flexibly estimated regression. DGP with cubic nonlinearity $q(x) = x^3$ and shock size $\delta = 1$. Each column corresponds to a sample size $T$; rows show the estimated ARF, mean bias, RMSE, and 90% coverage. Dashed: power series LP; solid blue: DR estimator pairing the same flexible regression with a parametric density ratio.

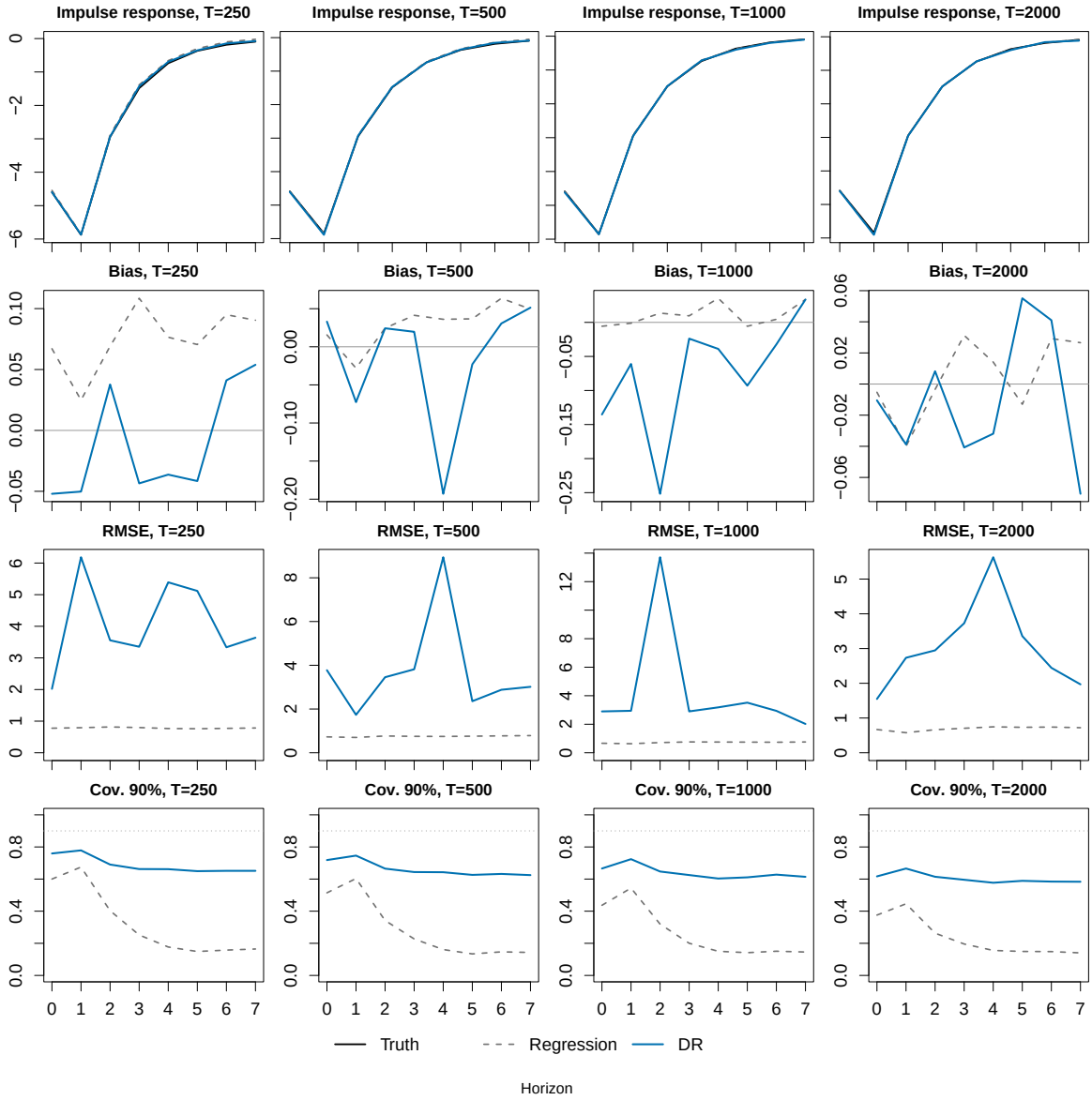


**Figure 8:** Flexibly estimated regression. DGP with cubic nonlinearity $q(x) = x^3$ and shock size $\delta = 2$. Each column corresponds to a sample size $T$; rows show the estimated ARF, mean bias, RMSE, and 90% coverage. Dashed: power series LP; solid blue: DR estimator pairing the same flexible regression with a parametric density ratio.